Proteins – from sequence to structure

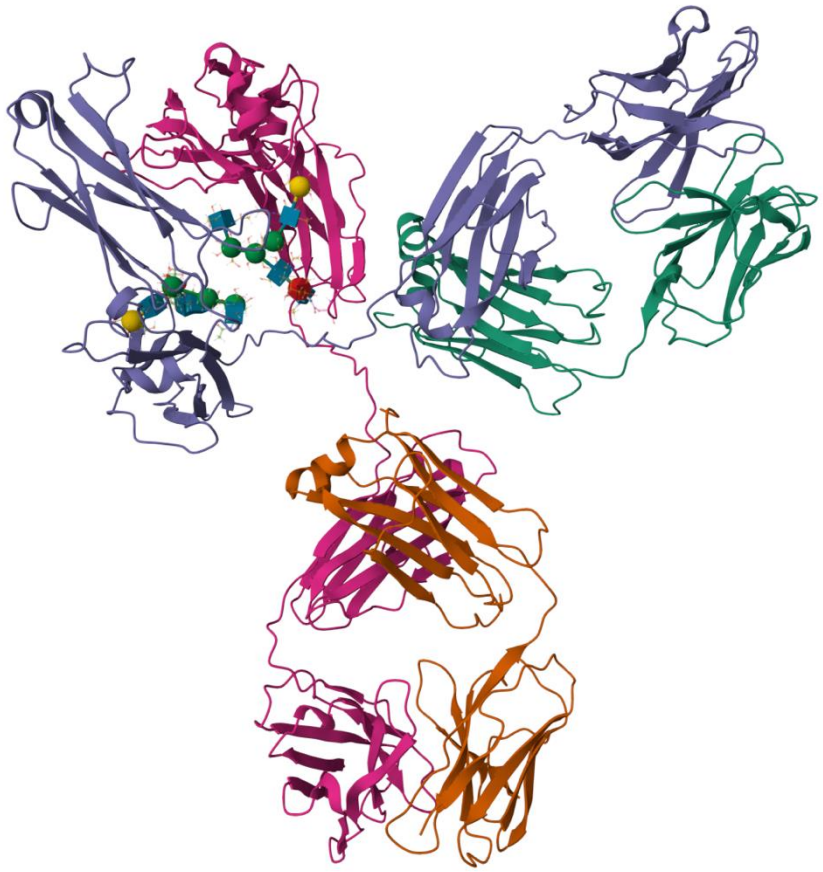
CEBT Part II

Advanced Biotechnology

Graham Christie

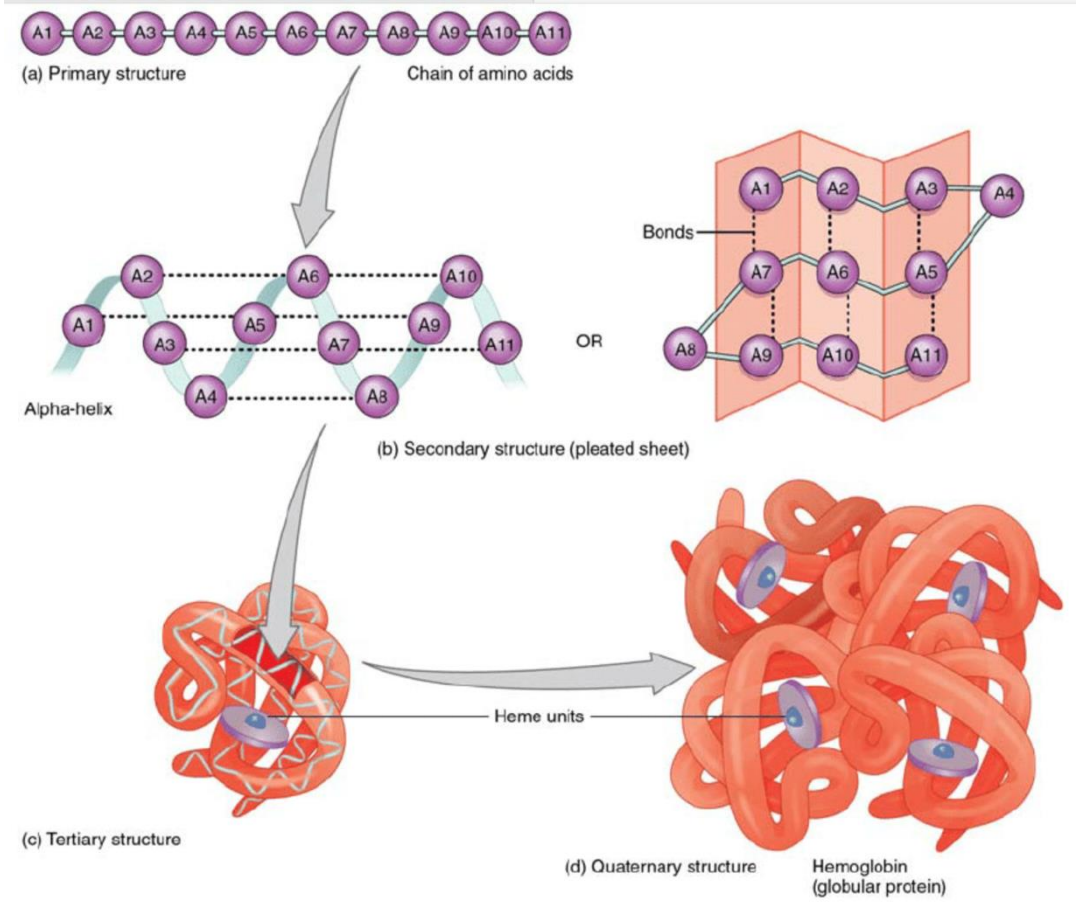
Proteins – why the interest?

- Major sector in biotechnology
  - Antibodies
  - Enzymes
  - Vaccines
  - Hormones
  - Insecticides
  - Food supplements etc
- Molecular machines that underpin all of life
- Next lecture: Protein Engineering



Proteins have four levels of organisation

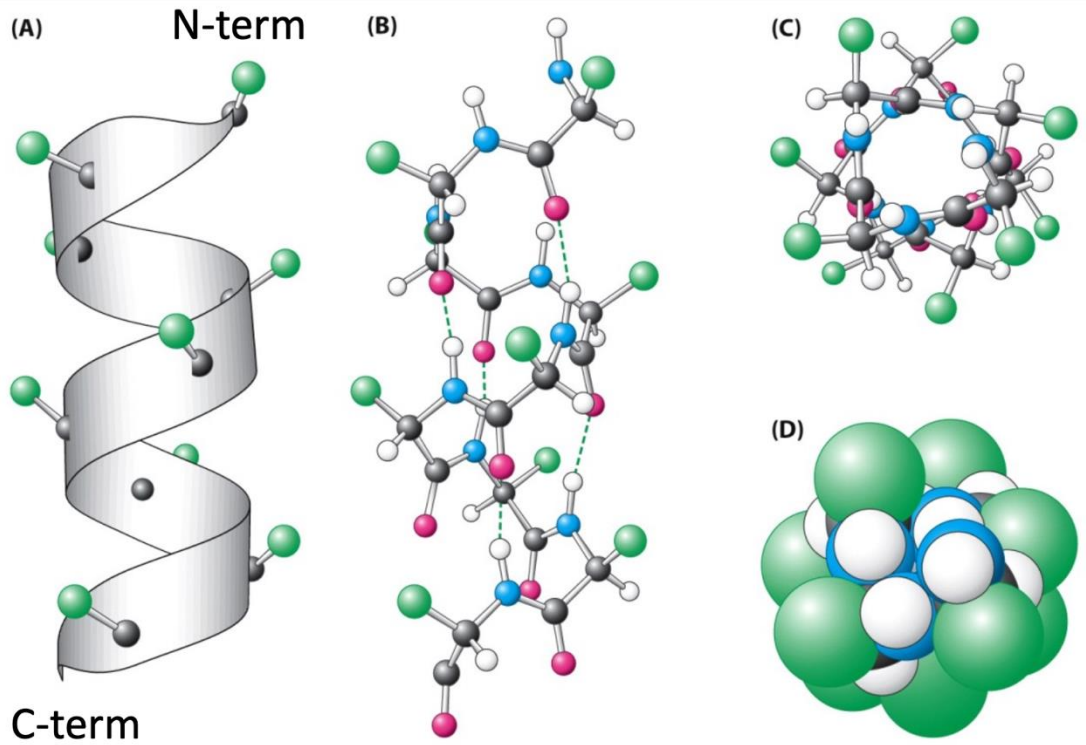
- Primary (amino acid sequence)
- Secondary - ‘local structures’ e.g. alpha helix and beta sheet
- Tertiary – folded polypeptide (*intra*molecular bonds), including any cofactors etc. Biologically active
- Quaternary – higher order assembly of discrete polypeptides (subunits; *inter*molecular bonds)



Protein secondary structure elements

Alpha Helix

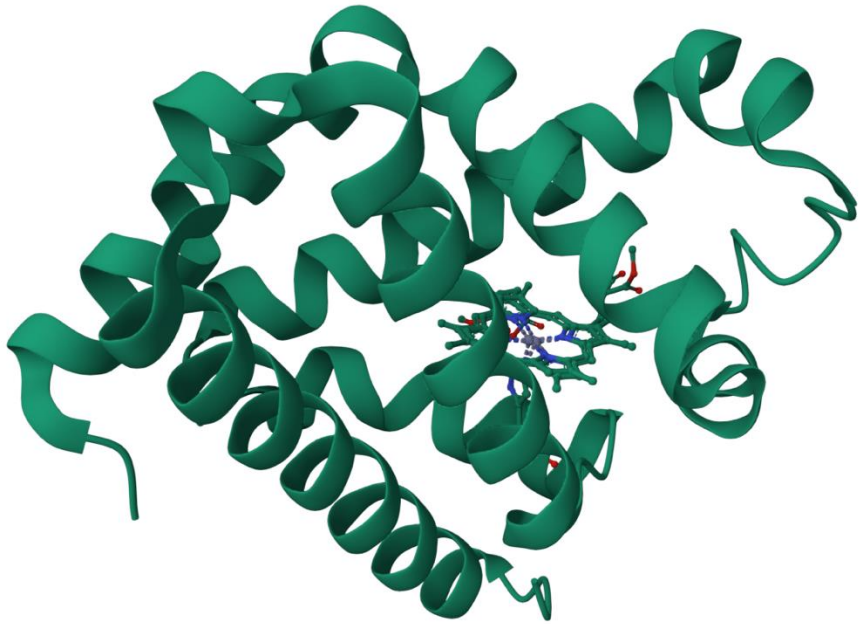
- Theorised by:
  - William Astbury (keratin)
  - Linus Pauling (paper models, X-rays)
- Characterised by:
  - ~10 – 40 amino acids (residues)
  - Right-handed helix, no space inside!
  - Hydrogen bonding between amide and carbonyl group 4 residues ahead (*i* + *4*)
  - 100° separation between residues (3.6 residues per turn)
  - Side chains extend outwards and downwards (Christmas tree) towards N-terminus



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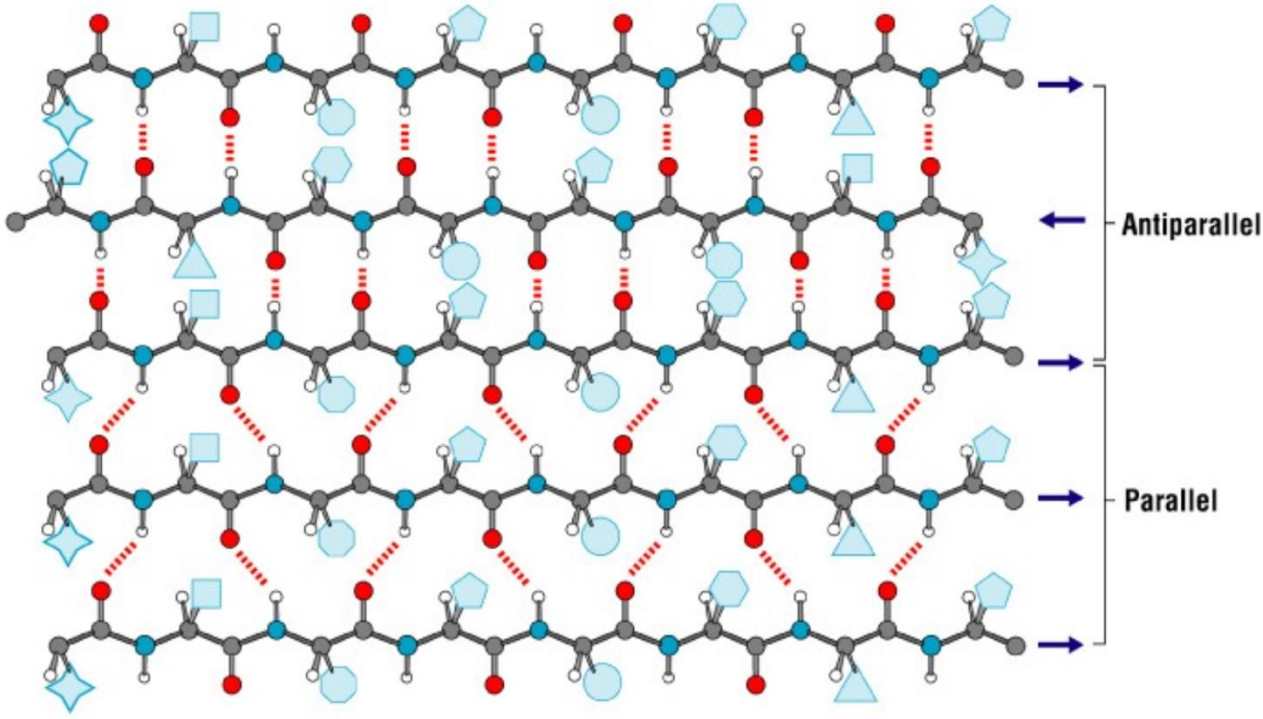


Myoglobin

Protein secondary structure elements

Beta (pleated) sheet

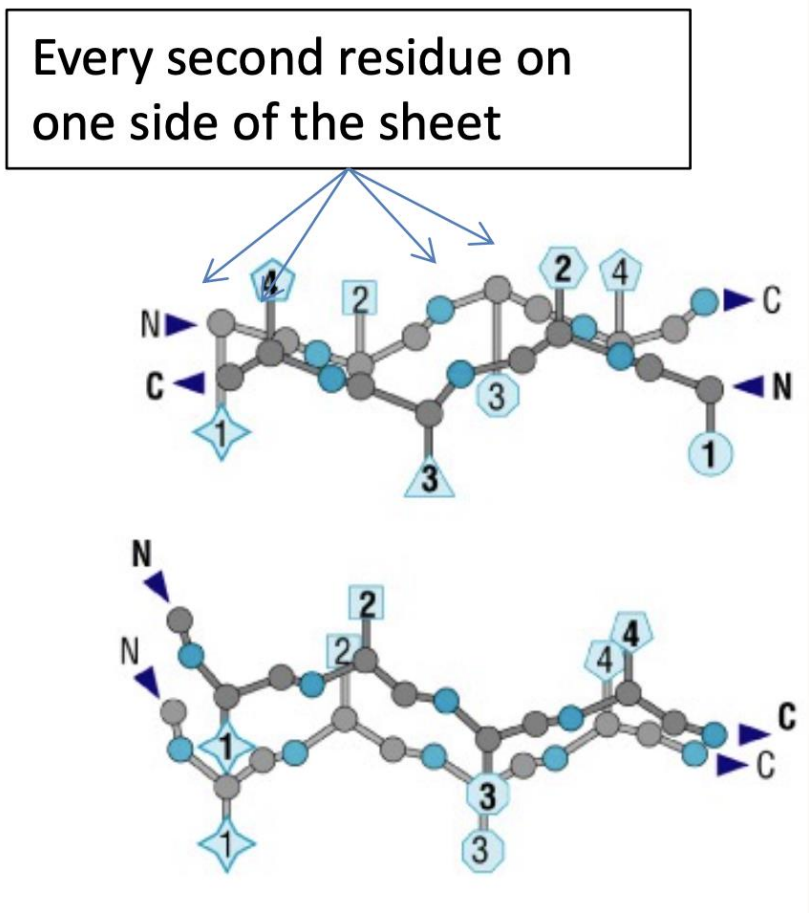
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- Characterised by:
  - Stretches of extended chains, β-strands
  - CO and NH form hydrogen bonds with neighbouring strands on both sides
  - Can be parallel or anti-parallel depending on direction of strands
  - Can form closed barrels e.g., GFP
  - Side chains point alternatingly in opposite directions along the strand but same direction across strands



Protein secondary structure elements

Beta (pleated) sheet

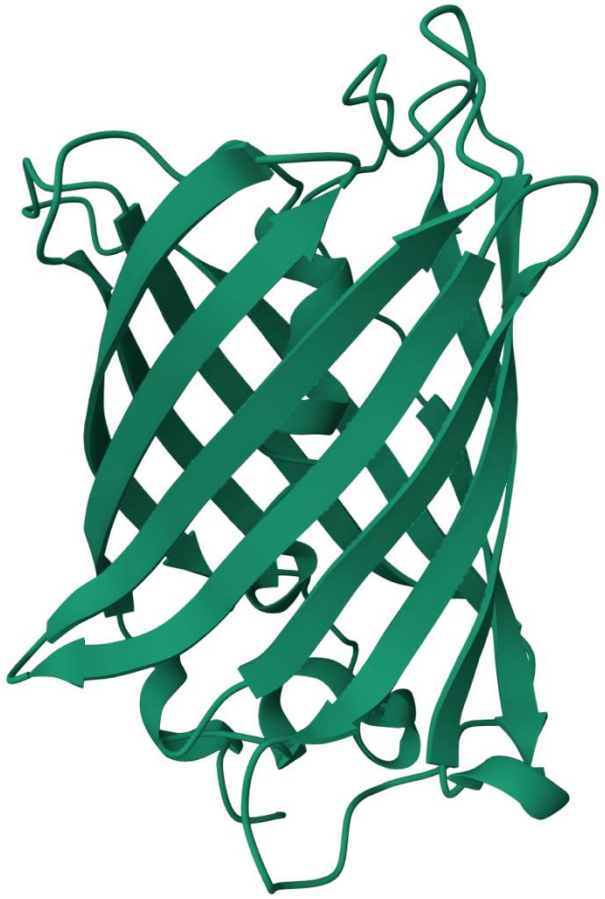
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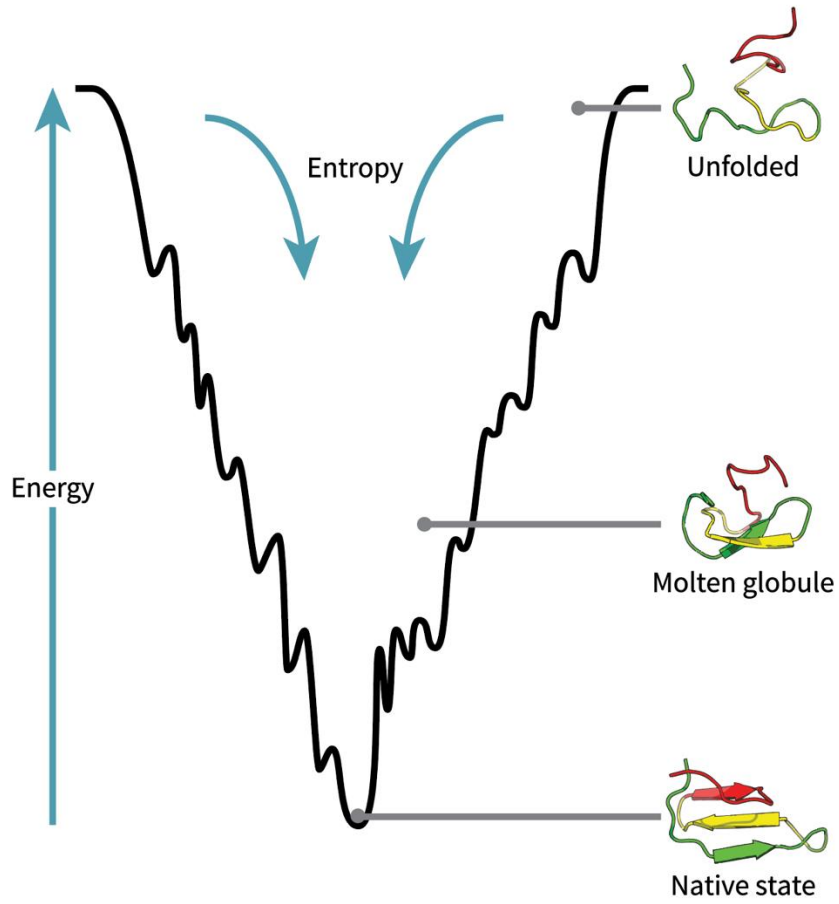
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Green Fluorescent Protein (GFP)

Protein folding

- Tertiary structure is defined by the amino acid sequence – no help required
- Levinthal – 100 residues, sample 3 Ramachandran angles = 3^{198} conformations
- Proteins fold in milliseconds!
- Native conformation is the most stable, but energy difference between folded and denatured small (20-40 kJ/mol)
- Hydrophobic effect critical but also hydrogen bonding, electrostatic etc.
- Folding follows limited number of pathways – ‘folding funnel’ energy landscape
- Intermediate state – molten globules (but not always)

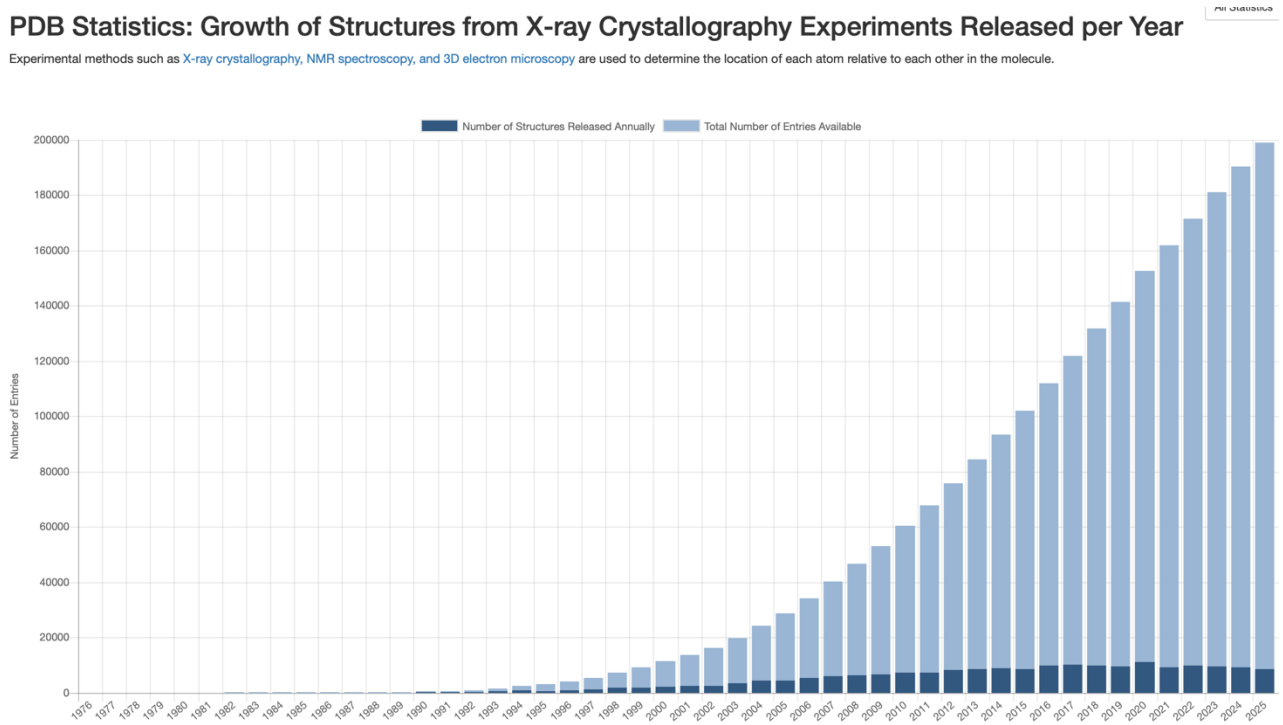


How to solve the structure of a protein

Experimental structure solution

Three principal methods

- X-ray crystallography
- Cryo Electron Microscopy
- NMR spectroscopy

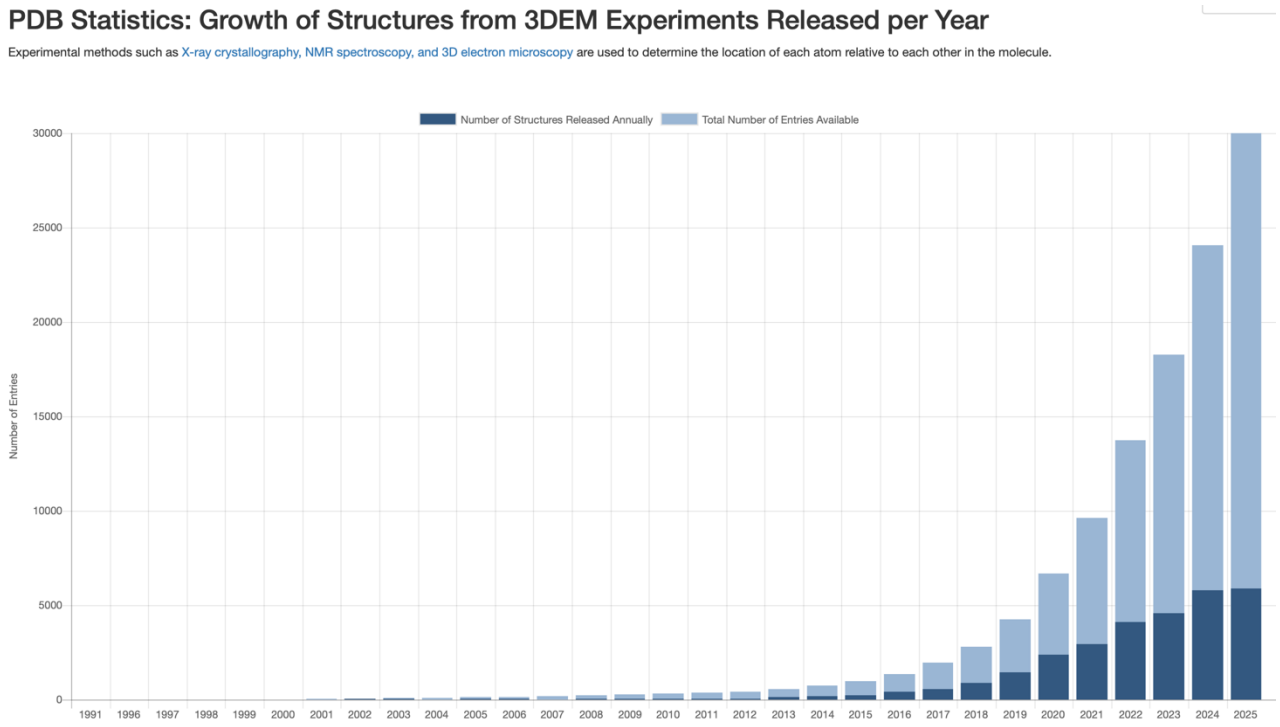


198,931 structures by X-ray crystallography

Experimental structure solution

<u>Three principal methods</u>

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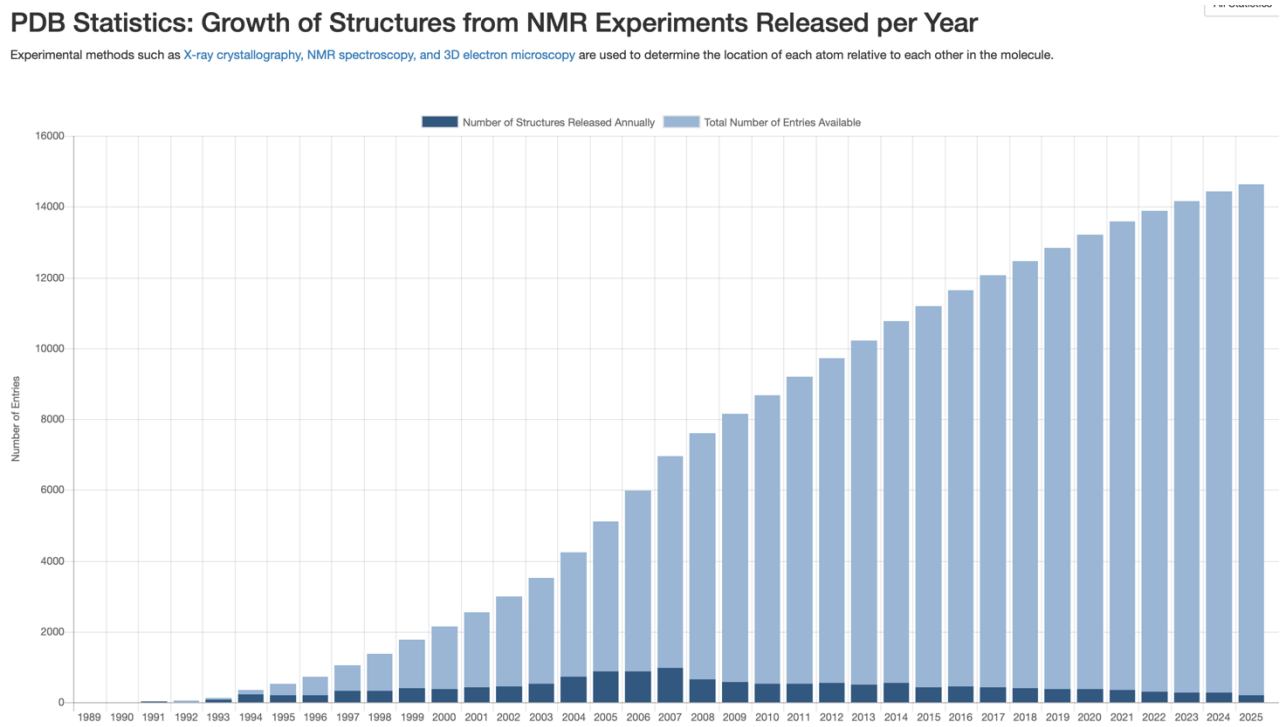


29,978 structures by EM

Experimental structure solution

Three principal methods

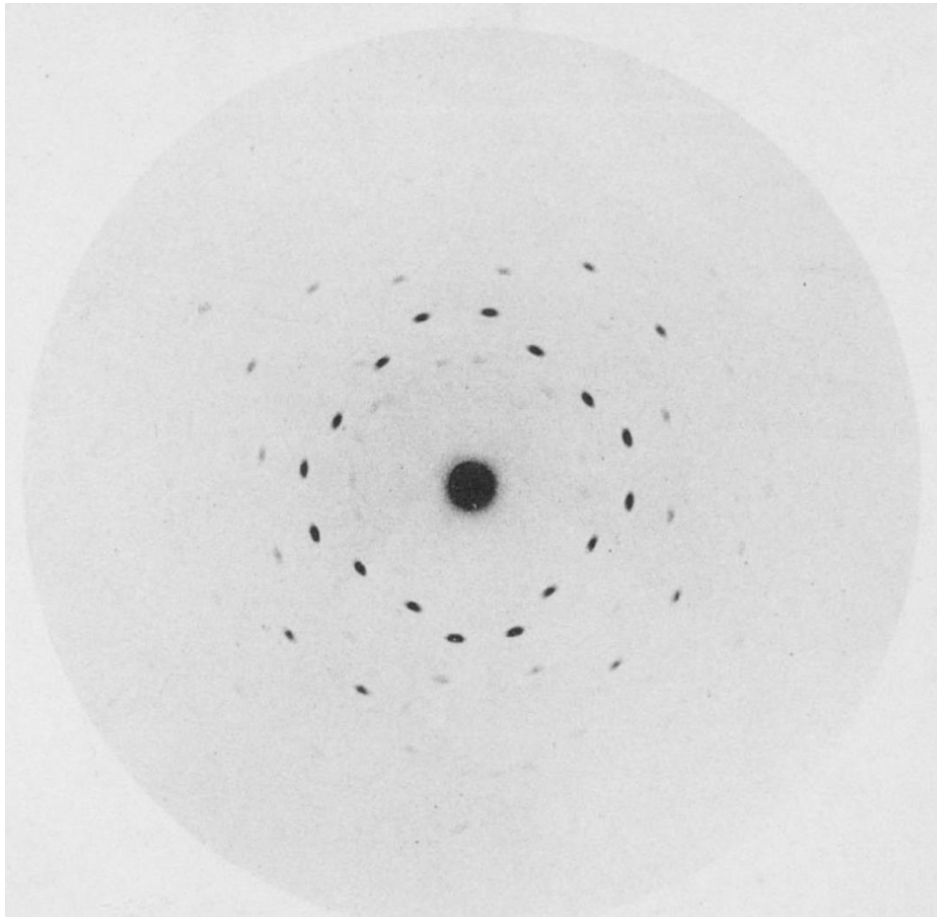
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14,623 structures by NMR

Macromolecular (X-ray) crystallography

- Max von Laue – Nobel Prize 1914 for discovery of diffraction of X-rays by crystals
- Lawrence Bragg "I can remember the exact spot on the Backs where the idea suddenly leapt into my mind that Laue's spots were due to the reflection of X-ray pulses by sheets of atoms in the crystal,"

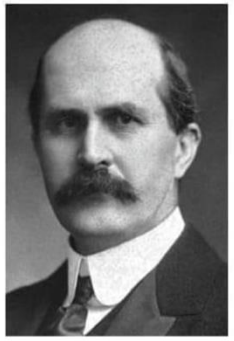


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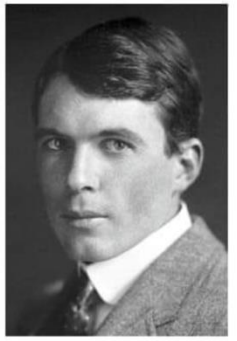
FATHER - SON COLLABORATION

The Nobel Prize in Physics 1915



Sir William Henry Bragg

Prize share: 1/2



William Lawrence Bragg

Prize share: 1/2

The Nobel Prize in Physics 1915 was awarded jointly to Sir William Henry Bragg and William Lawrence Bragg "**for their services in the analysis of crystal structure by means of X-rays.**"

In a letter to *Nature*, which continued to host much of the debate about the nature of X-rays, William referred readers to a paper in which **"my son has given a theory which makes it possible to calculate the positions of the spots for all dispositions of the crystal and photographic plate."**

Courtesy: Nobel Foundation

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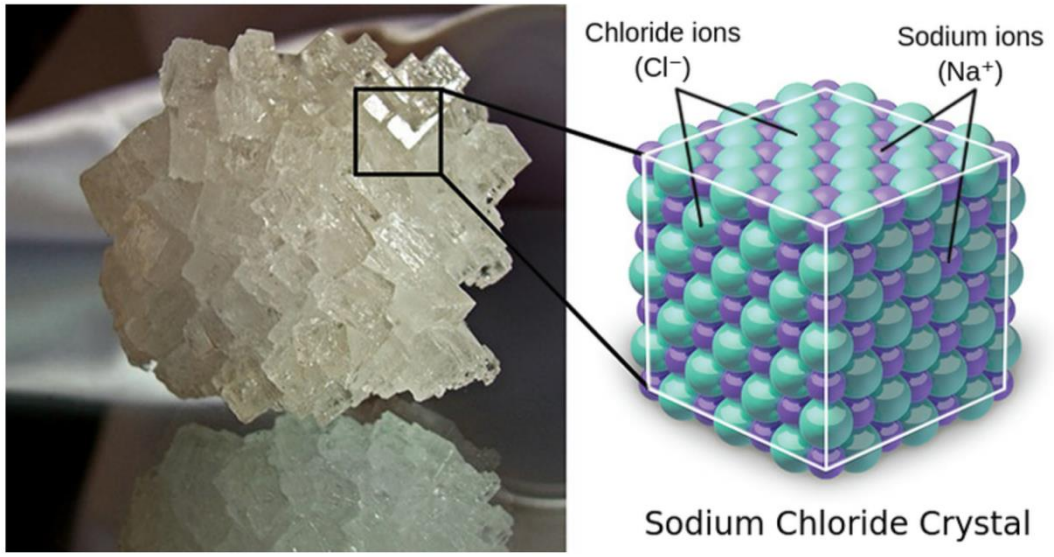
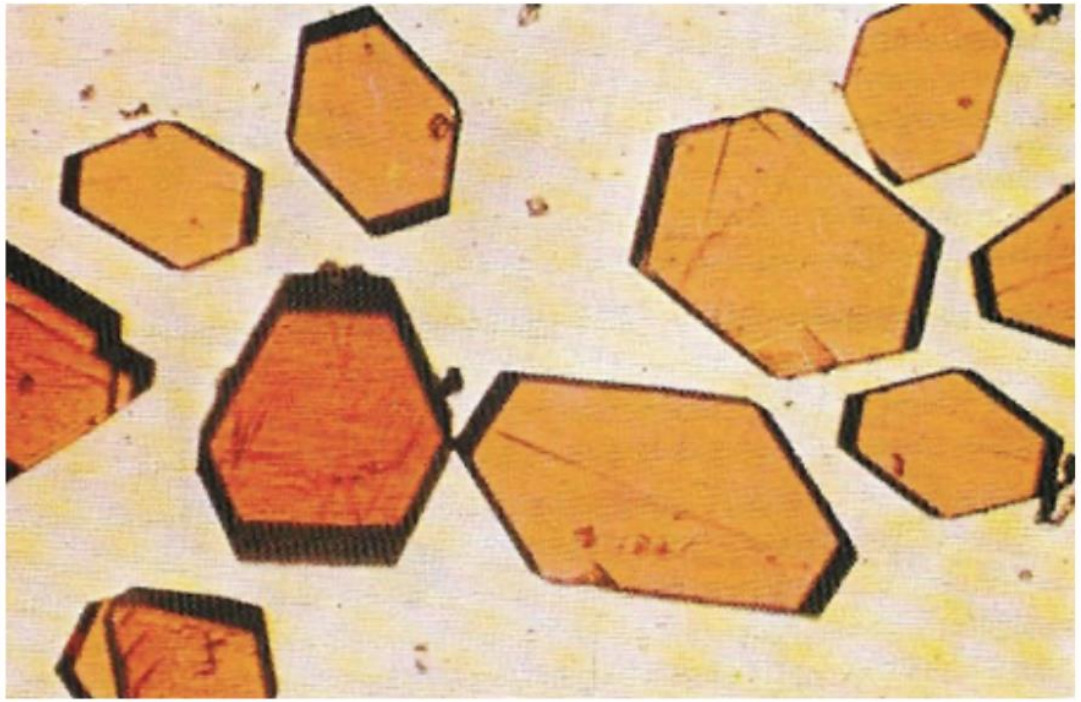
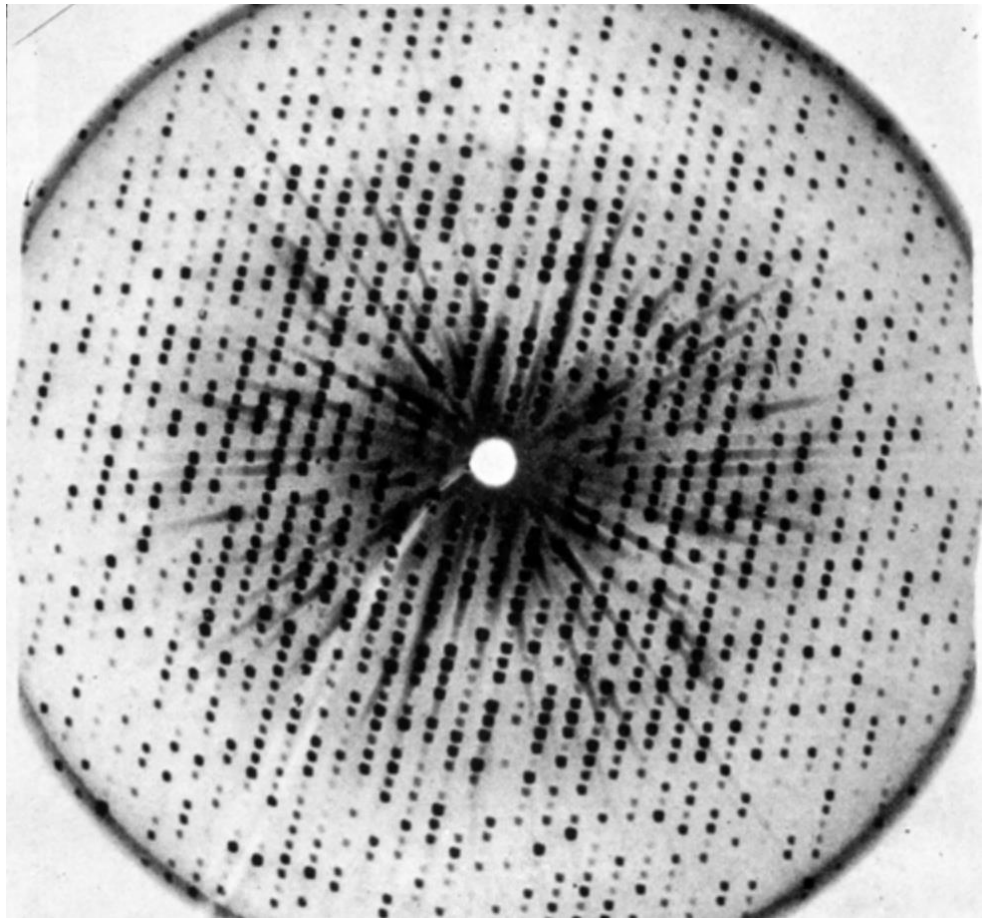


Figure 1: Crystal lattice of Sodium Chloride (NaCl) (Image by Blausen Medical Communications, Inc.)

Proteins can be crystallised and diffract X-rays

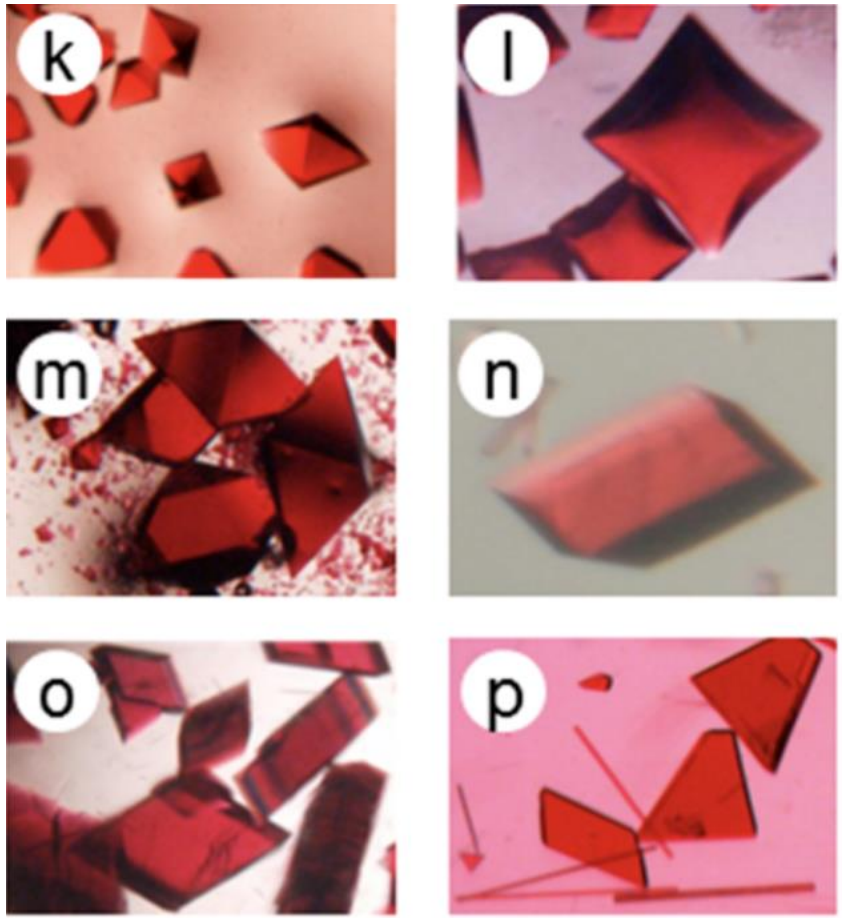


Myoglobin crystals

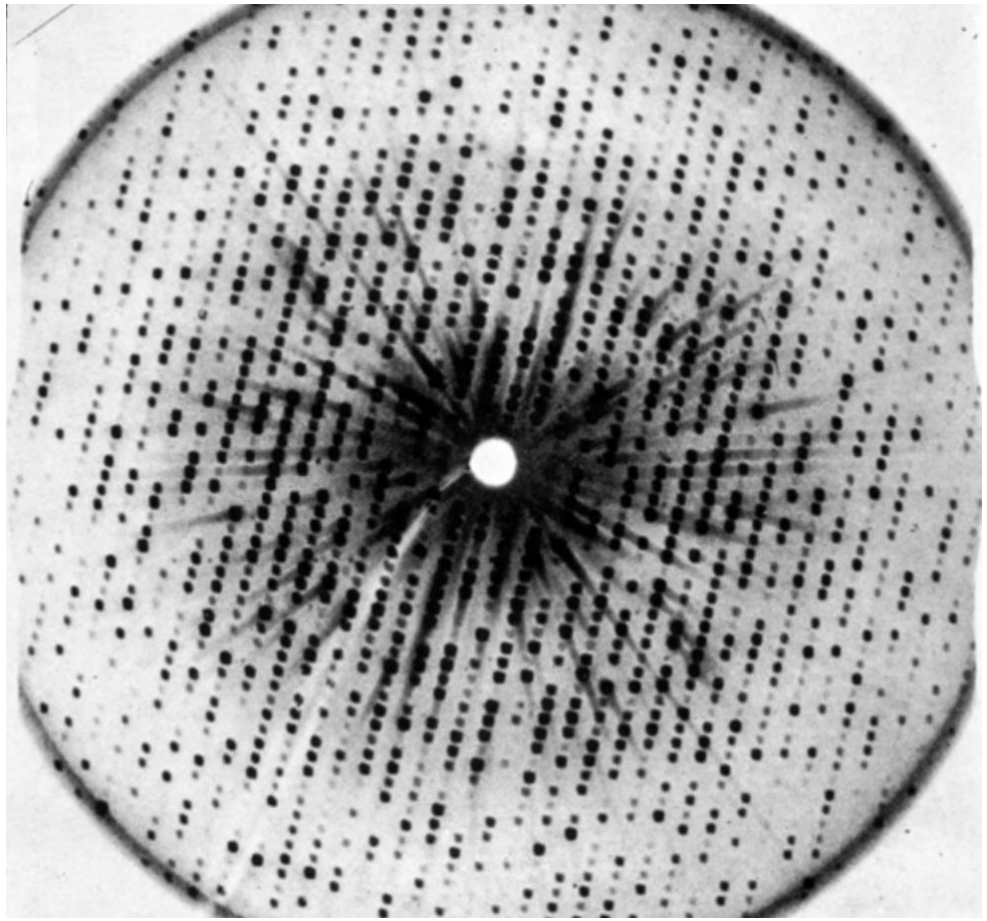


Myoglobin diffraction pattern

Proteins can be crystallised and diffract X-rays



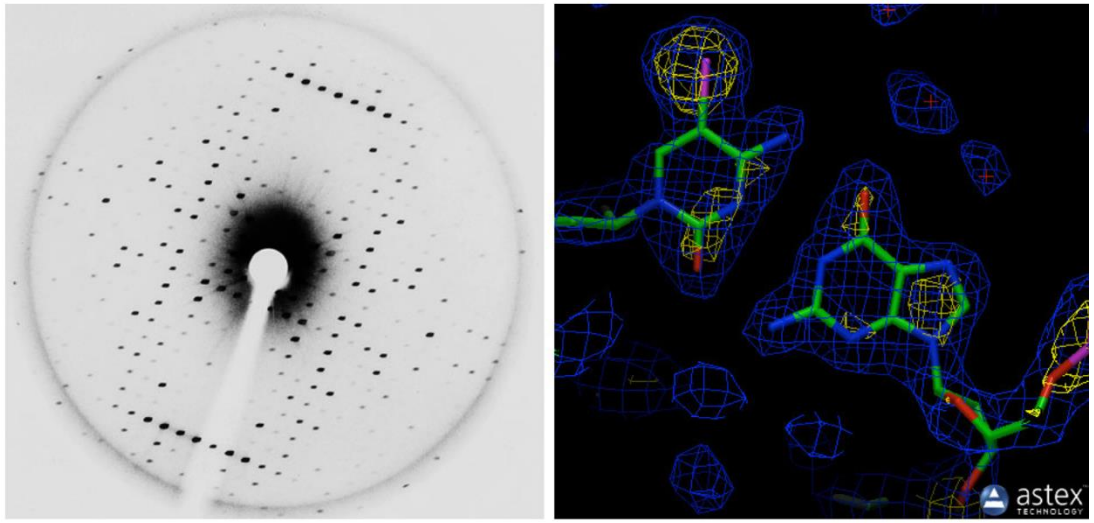
Haemoglobin crystals



Myoglobin diffraction pattern

Structure solution from diffracted X-rays

- Objective is to determine the arrangement of protein atoms in space
- Electron density in the unit cell (of the crystal) is related to the Fourier *transform* of the amplitude and phase of the scatted X-rays
- Reverse process - Fourier *synthesis* – can be used to reconstruct an electron density map of the unit cell
- Amplitude is square root of intensities of diffraction spots
- Phase information is lost – ‘the phase problem’



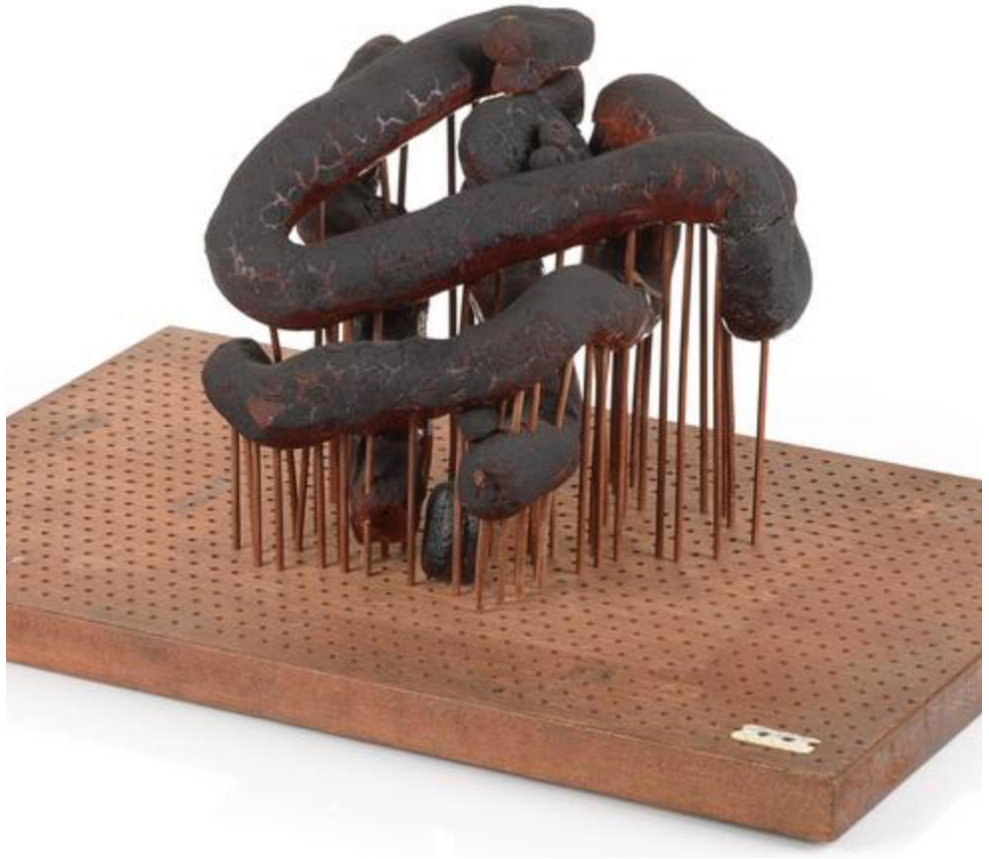
Structure solution from diffracted X-rays

- Max Perutz solved the phase problem by incorporating a few heavy atoms into haemoglobin crystals
- Result – a shift in the *intensity* of all diffraction spots
- Multiple Isomorphous Replacement permits retrieval of phase information
- John Kendrew – Perutz' PhD student – used the technique to solve the first protein structure (myoglobin, at 6 Å)
- Later extended to 2 Å - required computation for calculations



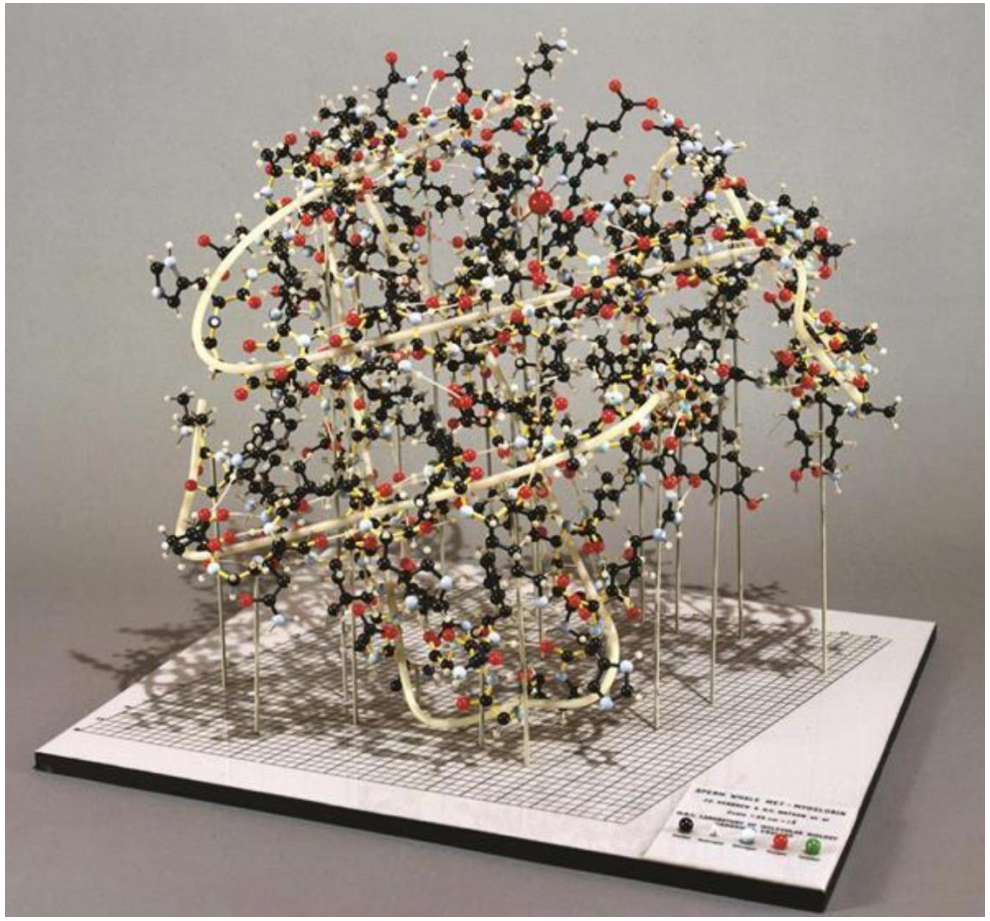
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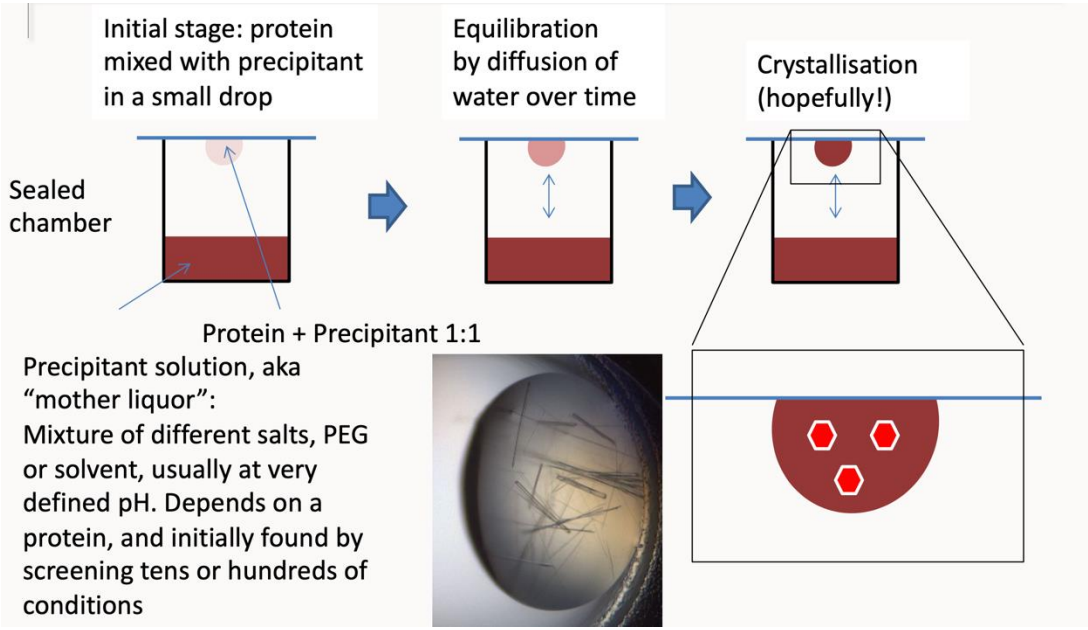
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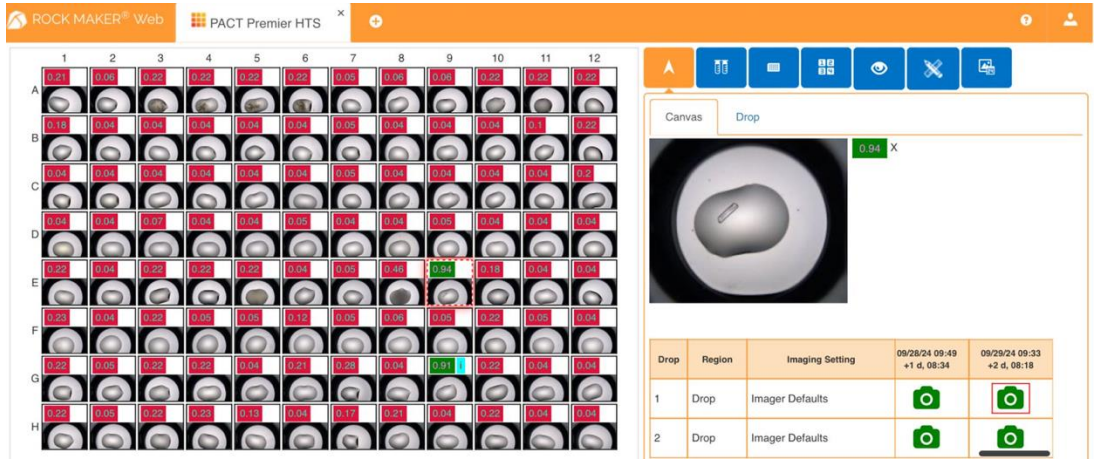
Modern X-ray crystallography workflow

- PCR and clone gene of interest
- Express recombinant protein in *E. coli* (or system of choice)
- Three stage protein purification:
  - Affinity (Ni-NTA for His_6-tag)
  - Ion exchange
  - Size exclusion chromatography
- Sitting (or hanging) drop crystallisation screens (nanolitre volumes, ~1000 conditions)
- Crystals fished, flash frozen, sent to Diamond synchrotron for data collection
- Phasing by Molecular Replacement (typically)



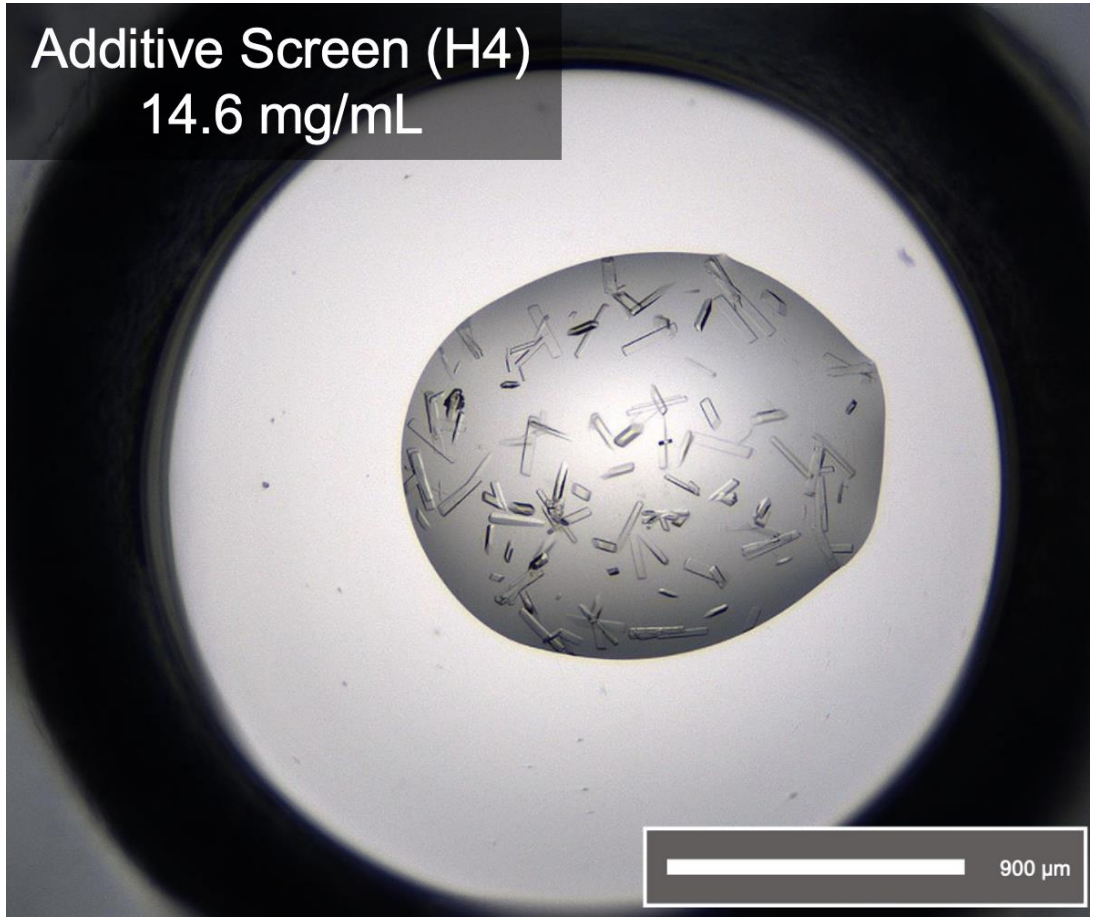
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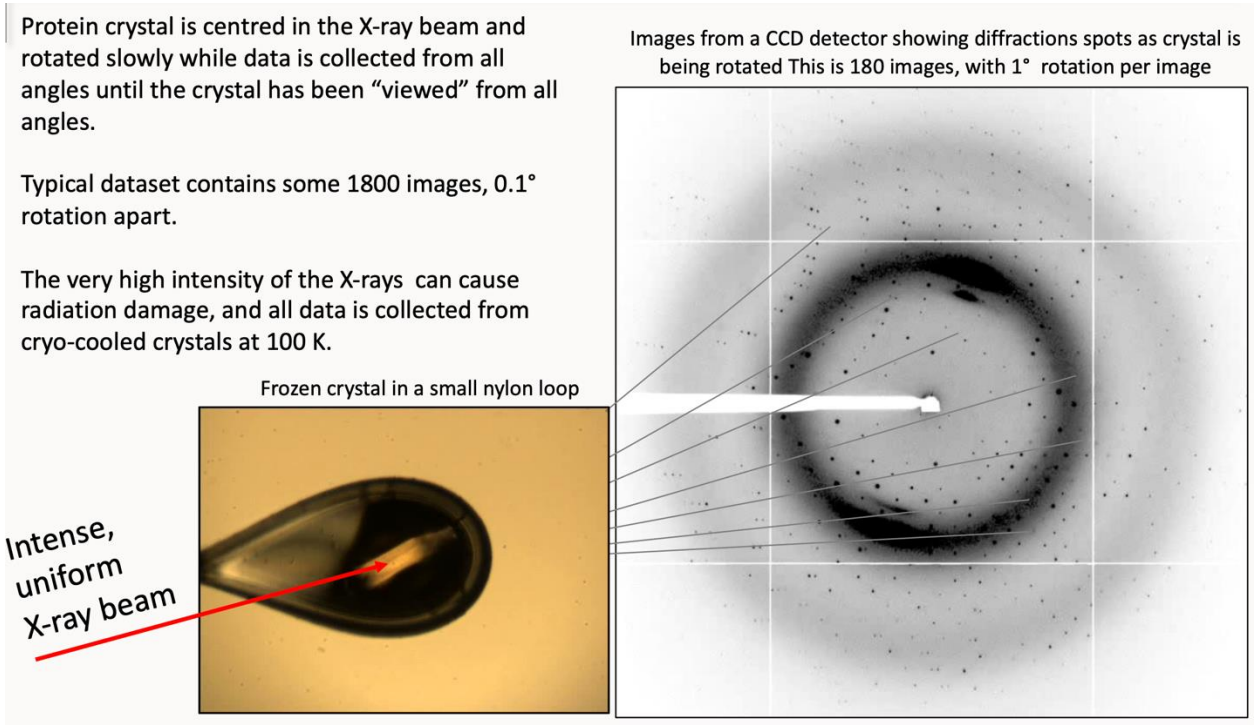
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Crystal is rotated to capture reflections from multiple angles i.e., to generate 3D electron density map

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Rotation anode X-ray in-house
-Single wavelength (CuKα = 1.54 Å)
- data collection in couple of hours

Modern X-ray crystallography workflow

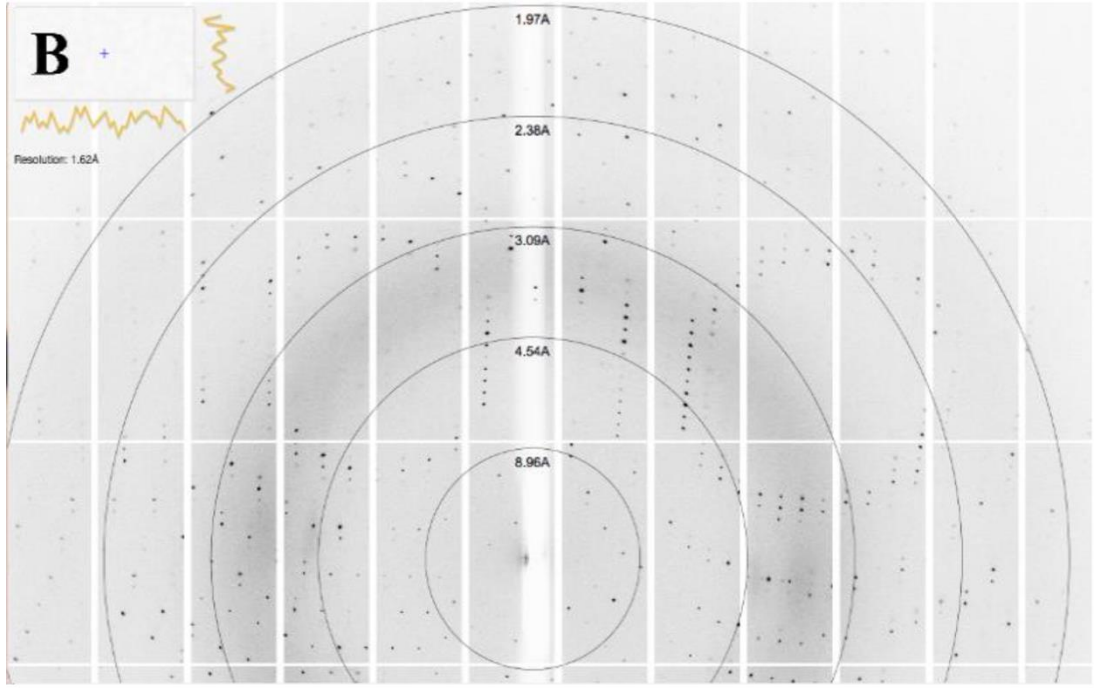
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Diamond Light Source

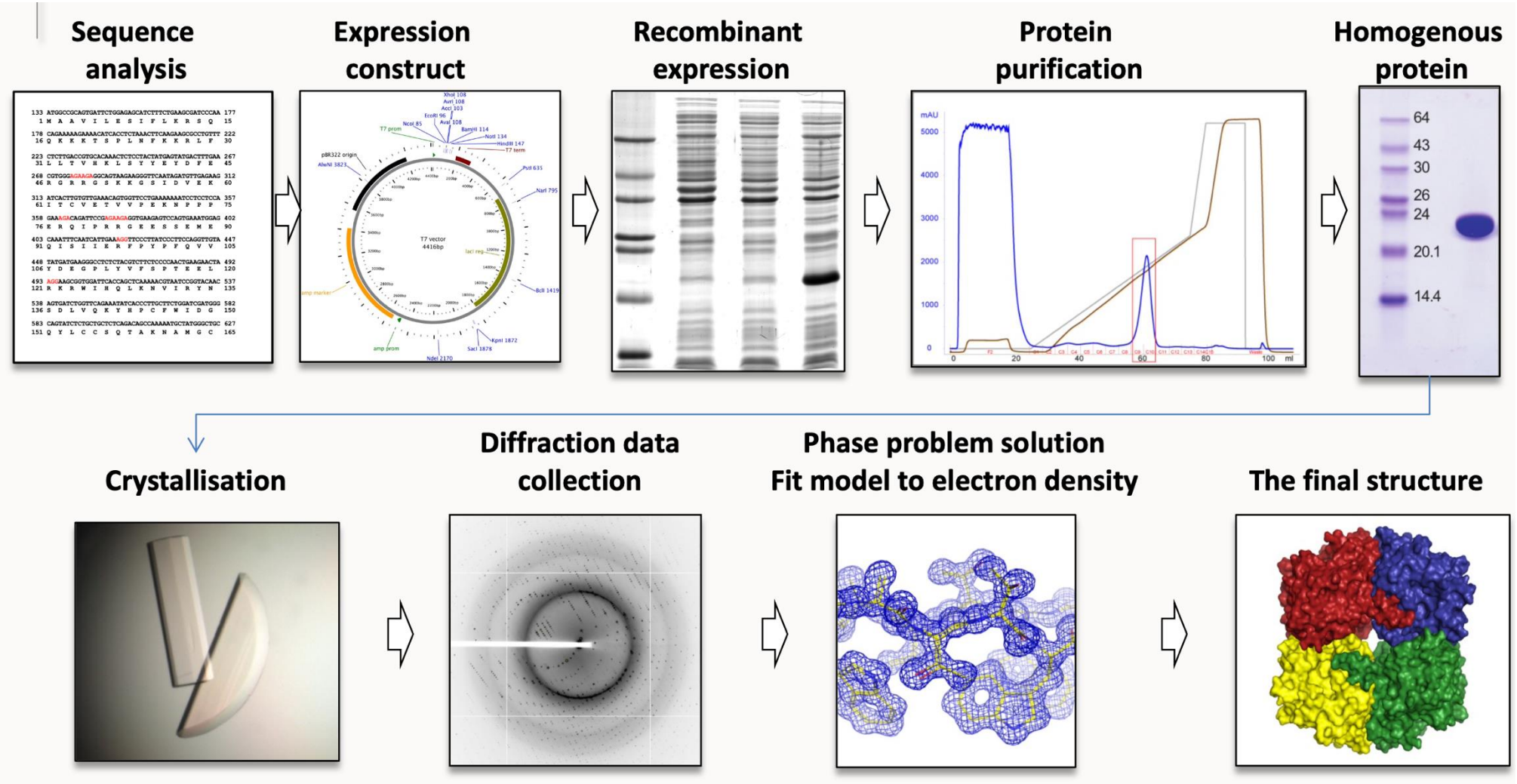
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Data collected to 1.8 Å (less with in-house X-ray source)

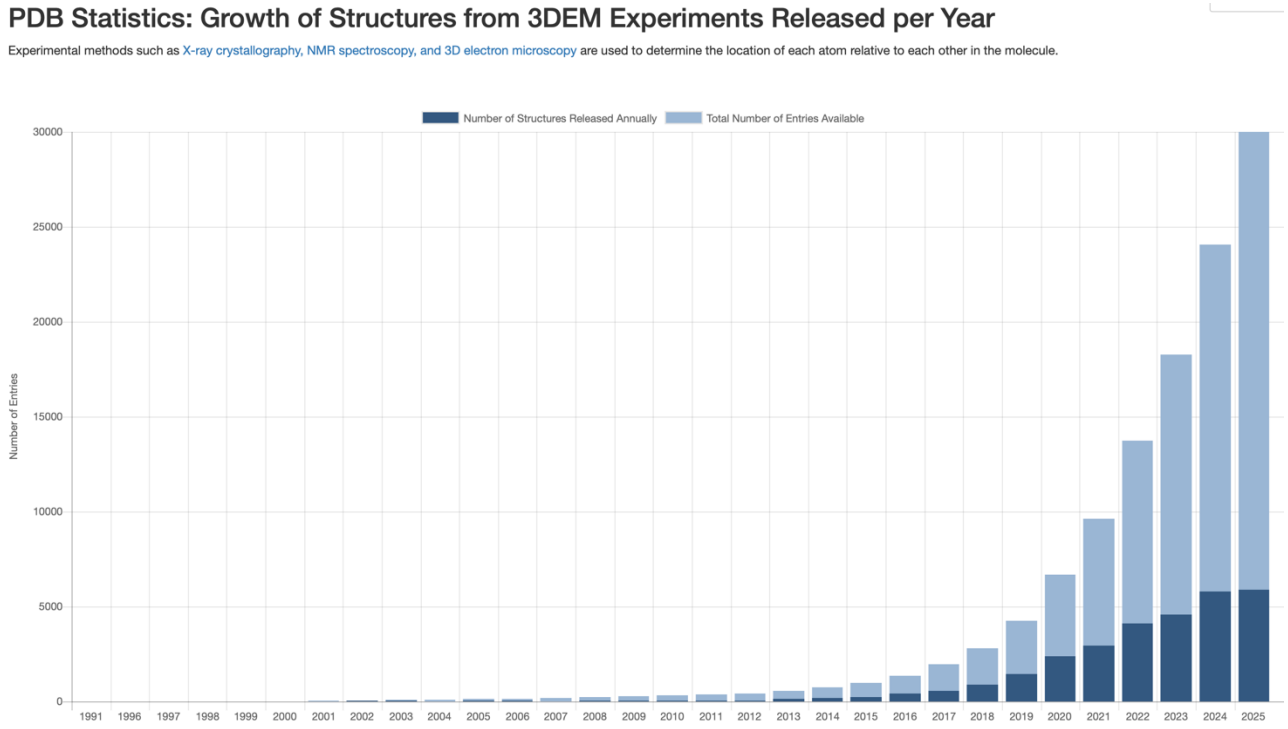
Protein crystallography summary



Experimental structure solution

Three principal methods

- X-ray crystallography
- Cryo Electron Microscopy
- NMR spectroscopy



29,978 structures by EM

Cryogenic Electron Microscopy (cryo-EM)

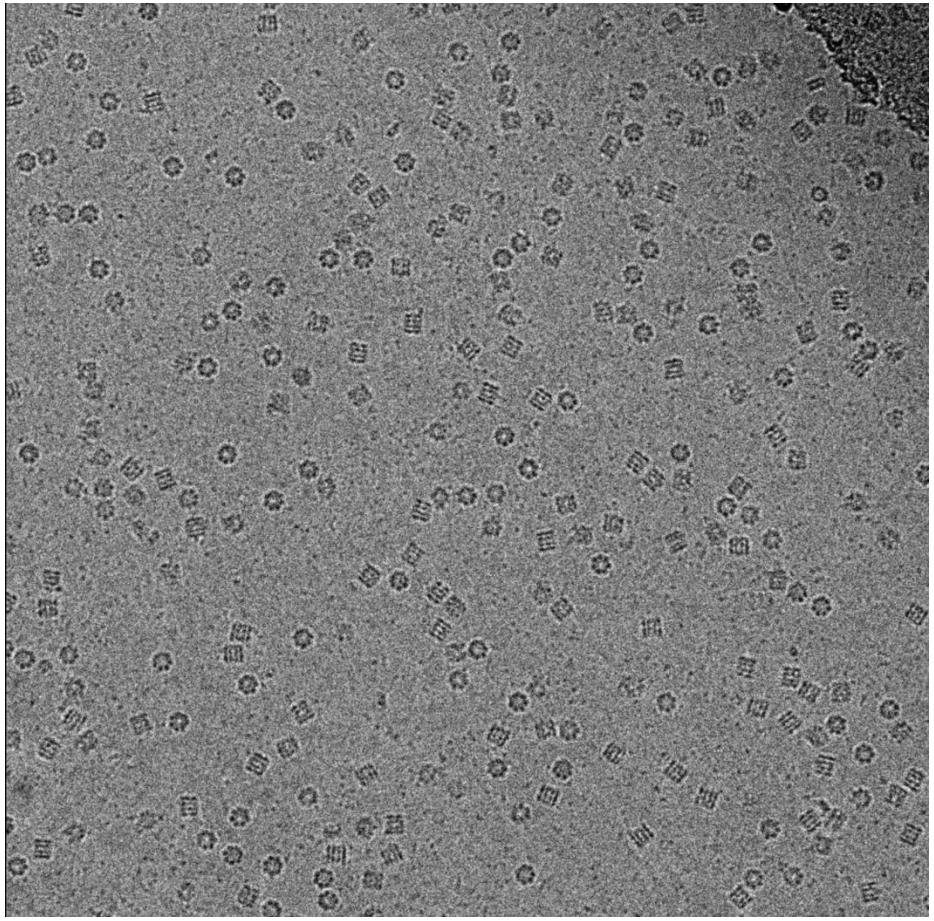
- Wavelength of electrons used for EM much shorter than visible light (0.02 Å versus ~500 nm)
- Limitation in resolution is molecular motion and beam-induced sample damage – blurring
- Standard TEM uses heavy metal stains (uranyl acetate etc) to minimise beam damage
- Cryo-EM method:
  - Samples vitrified in liquid ethane (-188C) – amorphous ice
  - Electron beam passes through sample ('see inside') generating 2D-projections
  - Thousands of 'particles' classified and averaged to improve resolution
  - Fourier transforms again but phase information is retained c.f., X-ray diffraction
  - EM density map of protein generated



Titan Krios EM

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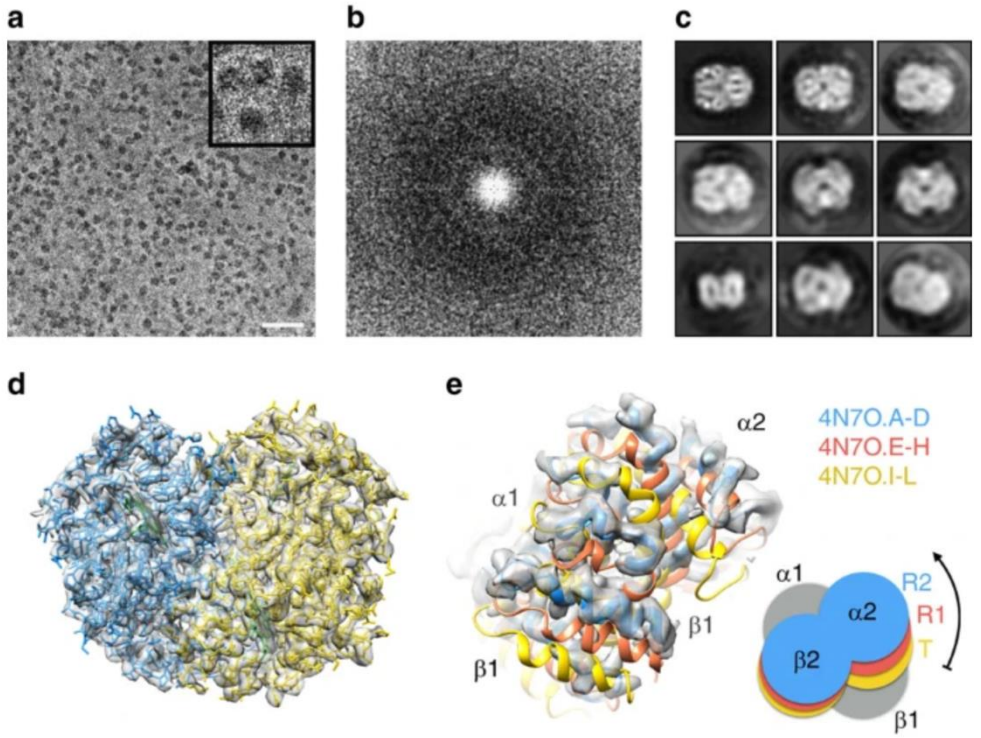


GroEL chaperone in amorphous ice at x50,000 mag

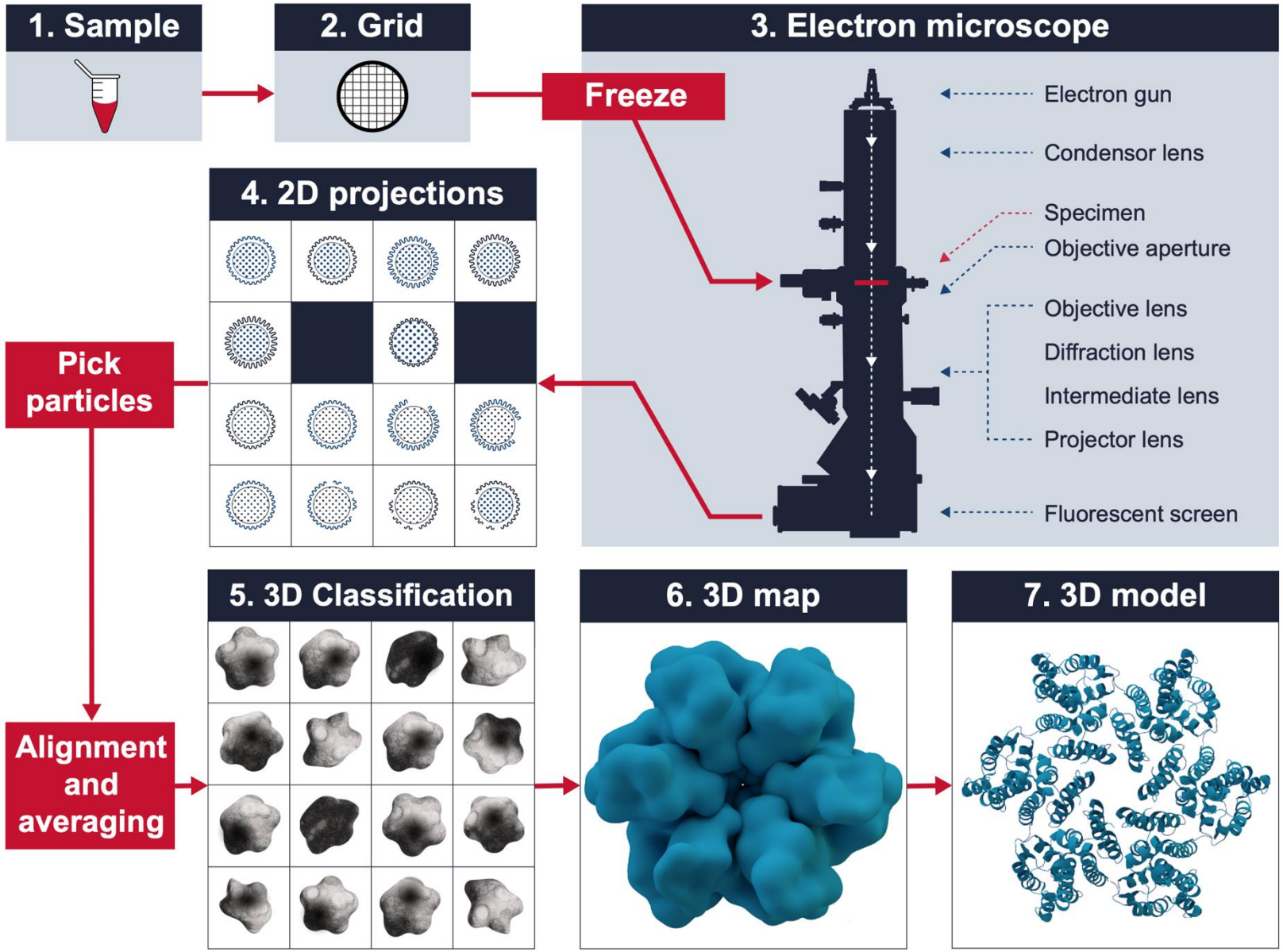
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  - EM density map of protein generated

Figure 1: Phase plate imaging of 64 kDa Hgb.



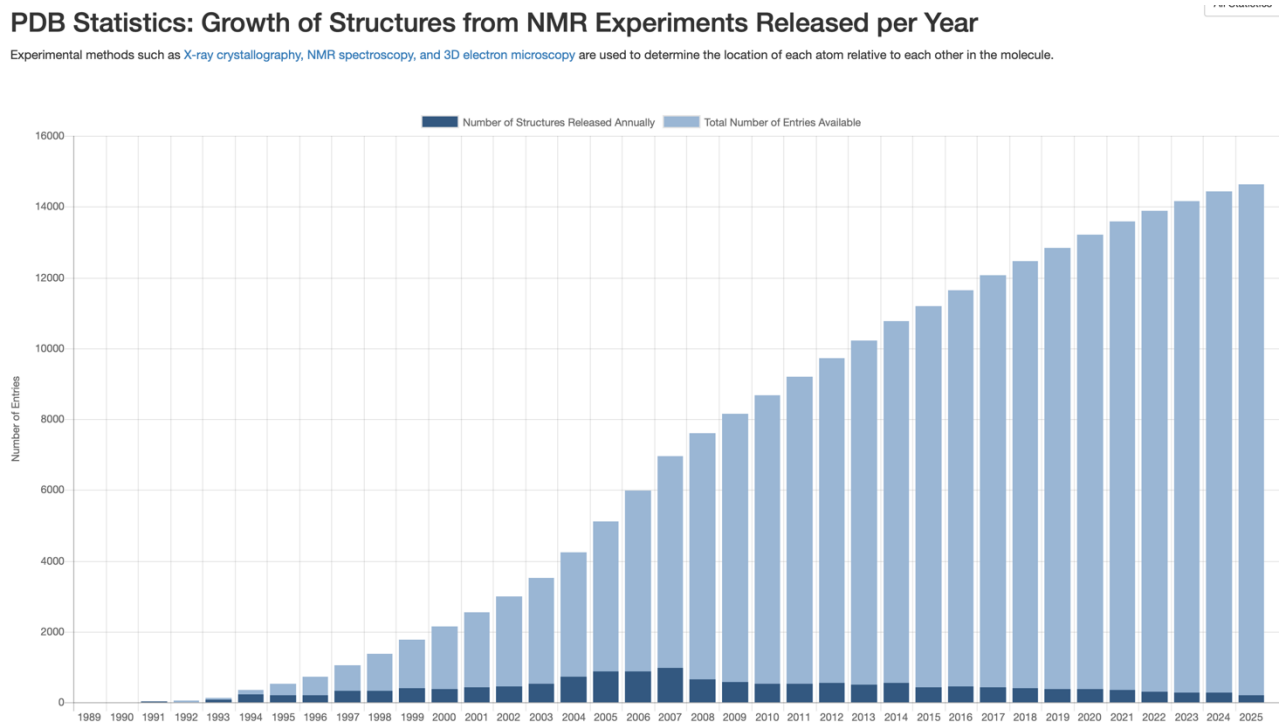
(**a**) Electron micrograph of metHgb recorded at ~500 nm underfocus with the Volta phase plate (VPP) (scale bar, 30 nm). (**b**) Power spectrum of the image in **a**, featuring contrast transfer function (CTF) Thon rings permitting defocus and phase shift estimation. (**c**) The 2D class averages of Hgb showing secondary structure elements in projection. (**d**) Reconstructed 3D electron scattering potential map and model of Hgb. (**e**) VPP reconstruction fitted with three conformers of Hgb present in crystal structure PDB 4N7O. The reconstructed 3D map agrees best with chains A–D of PDB 4N7O representing the R2 state of Hgb.


1. Sample
2. Grid
3. Electron microscope
Freeze
Electron gun
Condensor lens
Specimen
Objective aperture
Objective lens
Diffraction lens
Intermediate lens
Projector lens
Fluorescent screen
4. 2D projections
Pick particles
5. 3D Classification
Alignment and averaging
6. 3D map
7. 3D model

Experimental structure solution

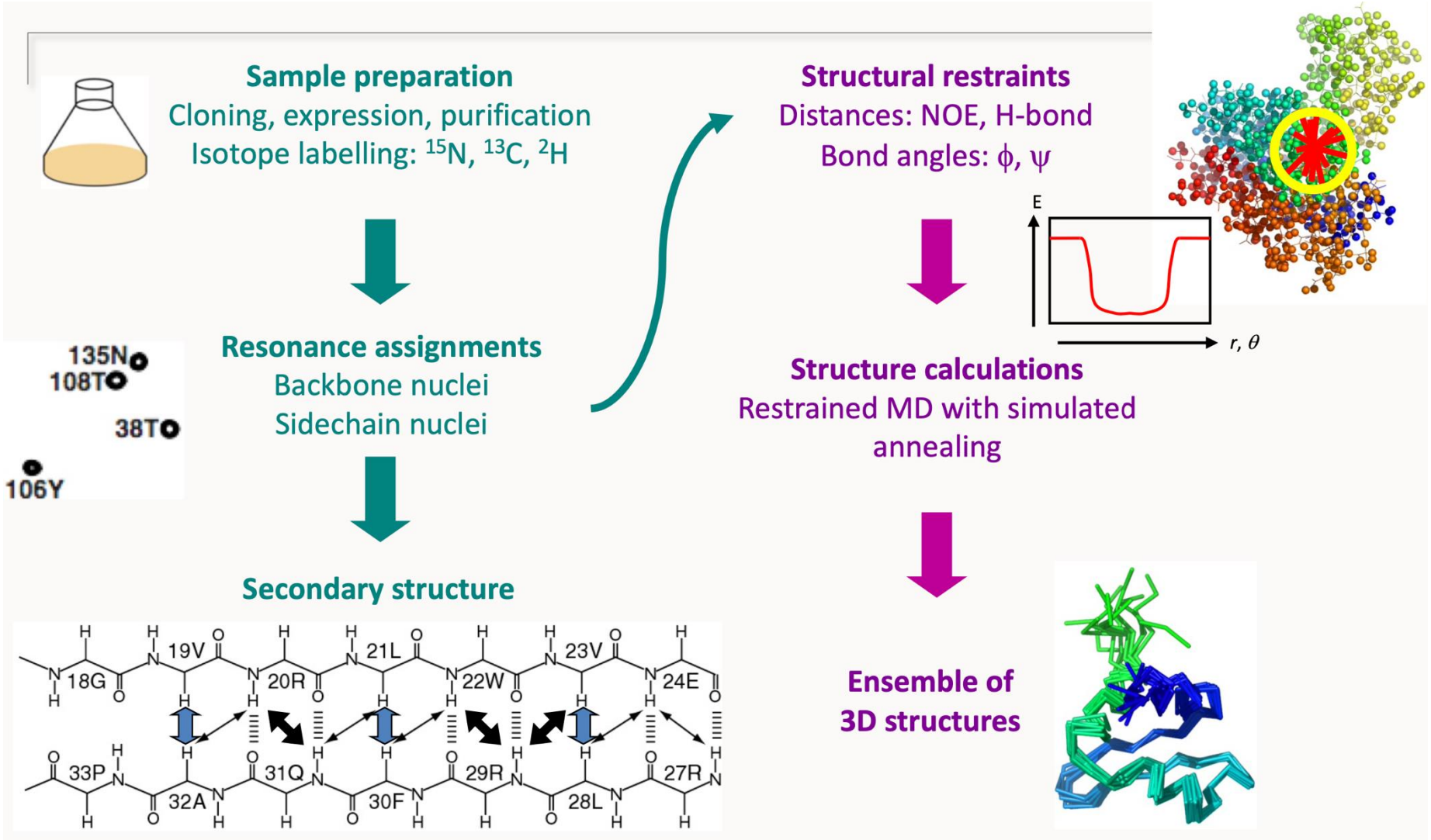
Three principal methods

- X-ray crystallography
- Cryo Electron Microscopy
- NMR spectroscopy



14,623 structures by NMR

NMR structure solution (for info only)



Experimental methods summary

X-ray crystallography

- Atomic resolution (0.5 – 2 Å)
- No size limitation – but proteins increasingly difficult to crystallise with size
- Bottlenecks:
  - mg quantities of homogenous protein
  - Crystal growth and quality
  - Membrane proteins challenging

Cryo-EM

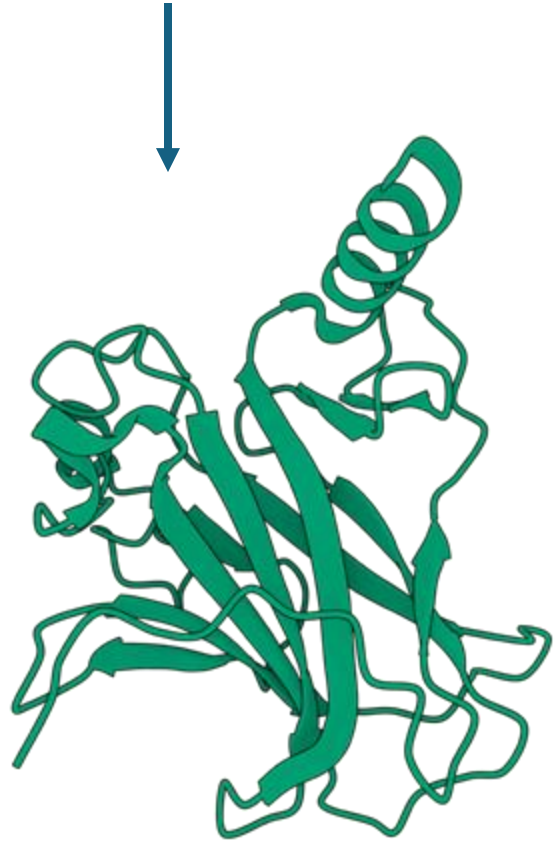
- Lower resolution (3-4 Å) but improving
- Better with larger proteins and protein complexes (>100 kDa)
- No need to crystallise
- Purity less important
- Proteins *in vivo* e.g., cryo-tomography

Computational methods for protein structure prediction

The protein folding problem…

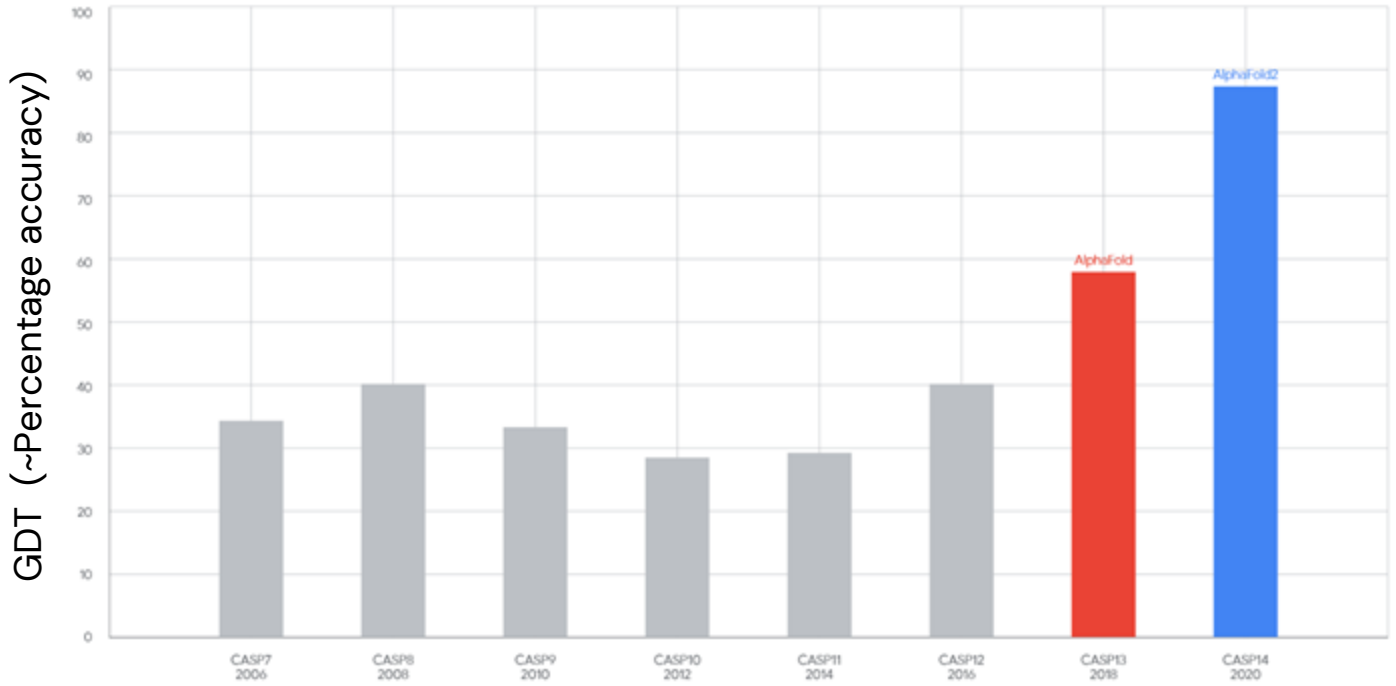
- Determining protein structure experimentally is inherently time and labour-intensive and involves many troubleshooting steps
- Can we predict protein 3D structure from primary sequence?
- **Anfinsen's dogma -** 3D structure of (most) natural proteins is determined only by its amino acid sequence
- This hypothesis was important, because it suggested we should be able to reliably predict a protein's structure from its sequence

…TAKSVTCTYSPALNKLFCQLAKTCPVQLWVDSTPPPGT
RVRAMAIYKQSQHMTEVVRRCPHHERCSDSDGLAPPQH
LIRVEGNLRAEYLDDRNTFRHSVVVPCEPPE…..



AlphaFold’s success in CASP

Critical Assessment of Structure Prediction (CASP) is a biennial experimental test of protein structure predictions

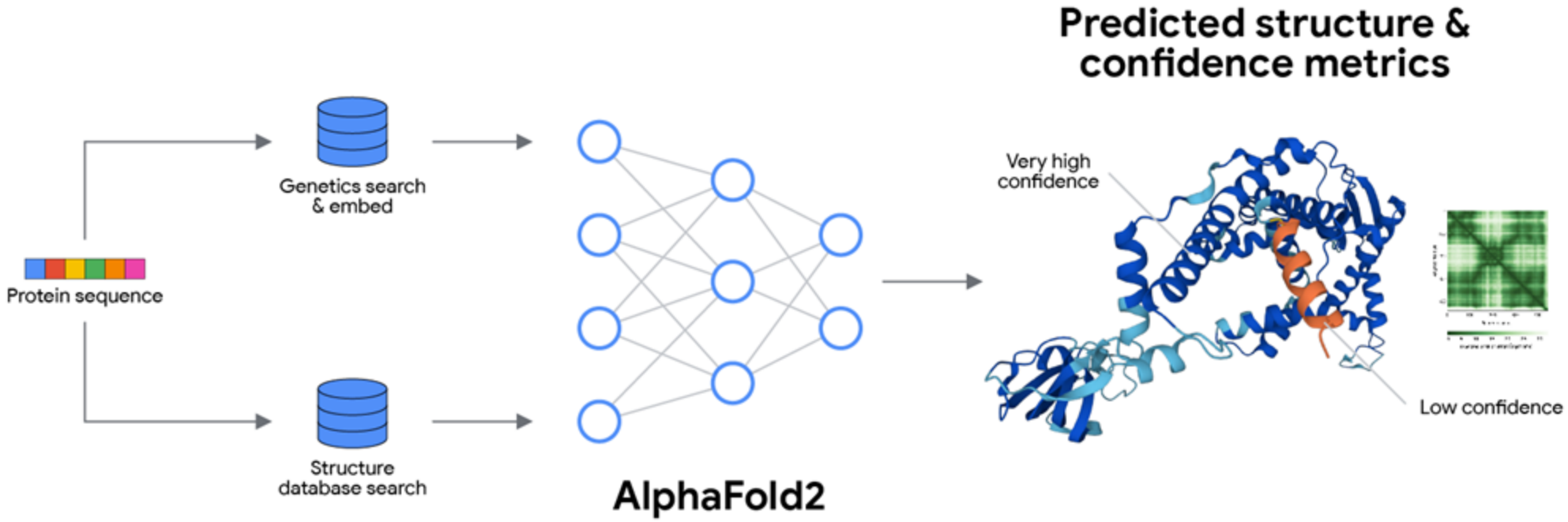


Overall success at protein structure prediction in CASPs over the years. AlphaFold drove rapid improvements in 2018 and 2020.

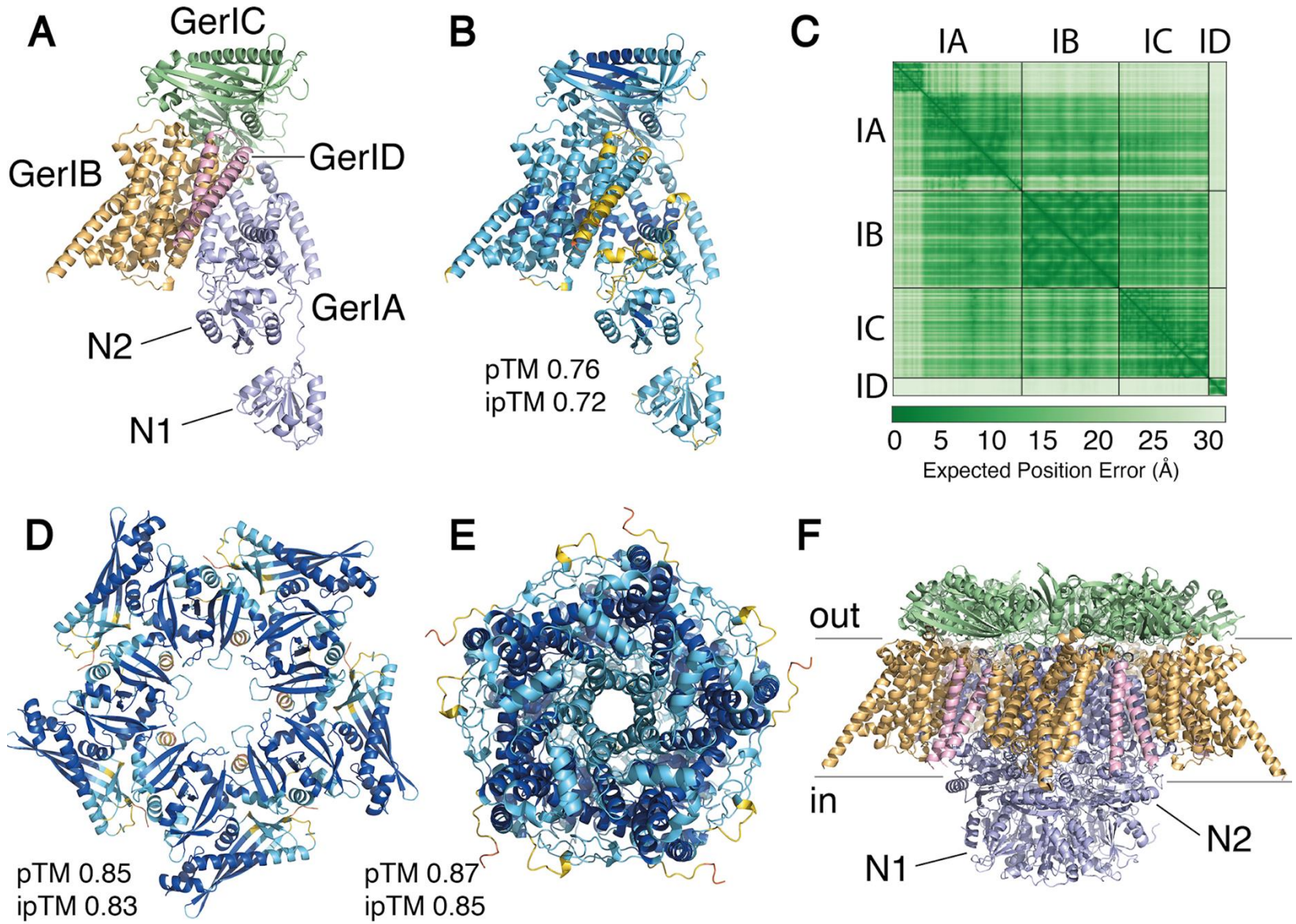
- Google DeepMind previously entered an earlier version of AlphaFold in 2018’s CASP13. It took the first place but by a small margin.
- Those predictions were not accurate enough, so the protein structure prediction problem was not considered solved.

AlphaFold2 is an advanced deep learning algorithm

- The training data for AlphaFold2 came from the Protein Data Bank (PDB)
- Crucially, AlphaFold also provides confidence metrics, such as pLDDT, pTM and PAE. If it is not sure about some part of the structure, it will tell you this in the form of a confidence score, enabling critical interpretation



Jumper, J et al. Highly accurate protein structure prediction with AlphaFold. Nature (2021)



Google DeepMind

PDBe Protein Data Bank in Europe EMBL-EBI

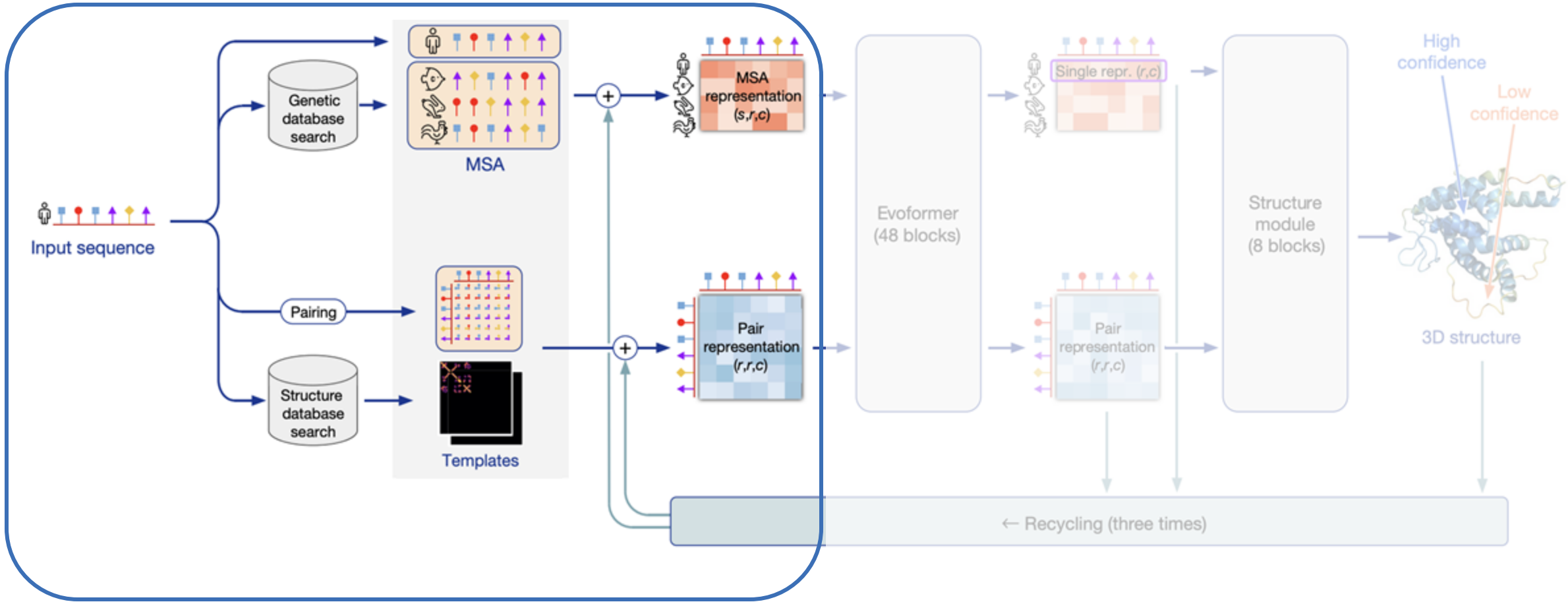
Strengths and limitations of AlphaFold 2

AlphaFold2 predicts	AlphaFold2 struggles to predict	AlphaFold2 doesn't predict
• Single protein chains • Protein multimers • Multisubunit protein-protein complexes	• Multiple conformations for the same sequence • Effects of point mutations • Antigen-antibody interactions	• Protein-DNA and protein-RNA complexes • Nucleic acid structure • Ligand and ion binding • Post-translational modifications • Membrane plane for transmembrane domains

AlphaFold: High-level overview



Inputs



Google DeepMind

PDBe Protein Data Bank in Europe EMBL-EBI

Jumper, J et al. Highly accurate protein structure prediction with AlphaFold. Nature (2021).

The role of multiple sequence alignment (MSA)

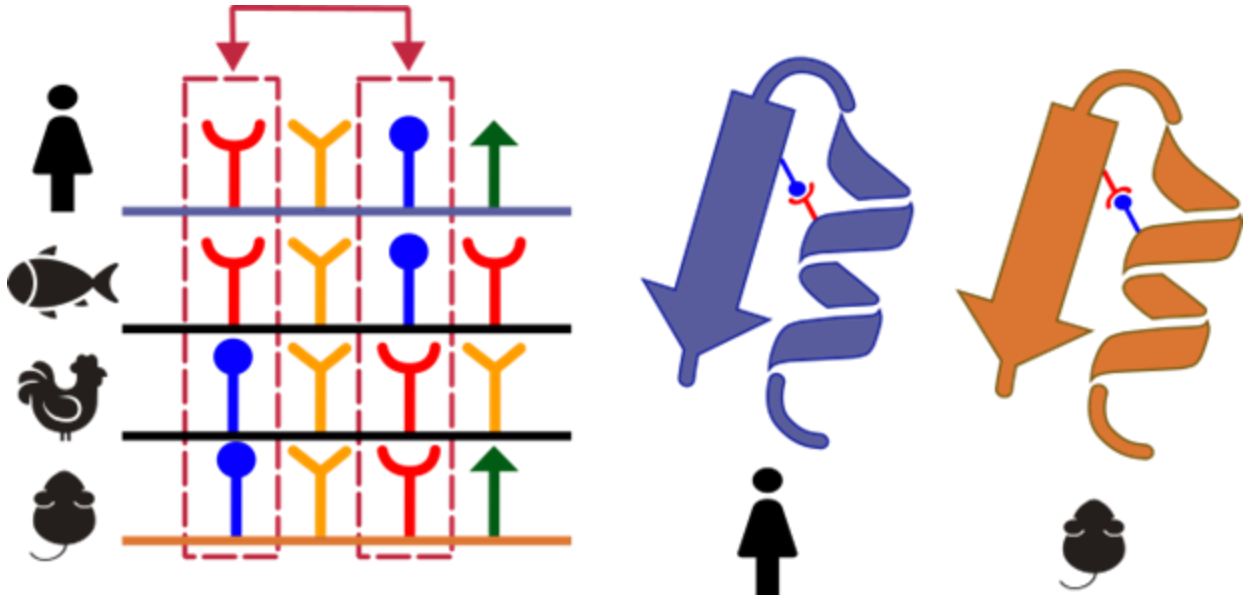
- The primary input for AlphaFold2's neural network is then the MSA.
- AlphaFold2 uses MSAs to compare and analyse the sequences of similar proteins from different organisms. It highlights similarities and differences, which helps understand the evolutionary relationships between the proteins.



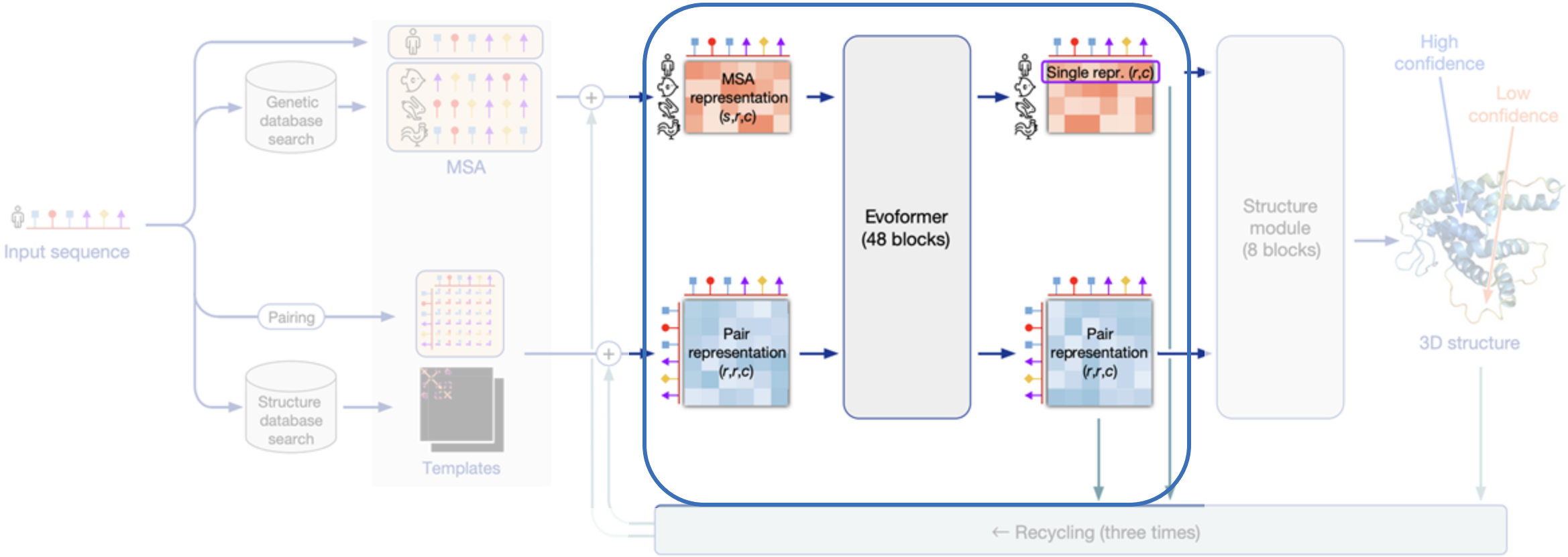
Co-evolution

If two amino acids in a protein are in close contact, mutations in one of them will probably be followed by mutations of the other. This preserves the structure of the protein, and is known as co-evolution or covariation.

The opposite is also true: if two regions of a protein are changing and evolving independently from each other, it is likely that they are not in direct contact.



Evoformer



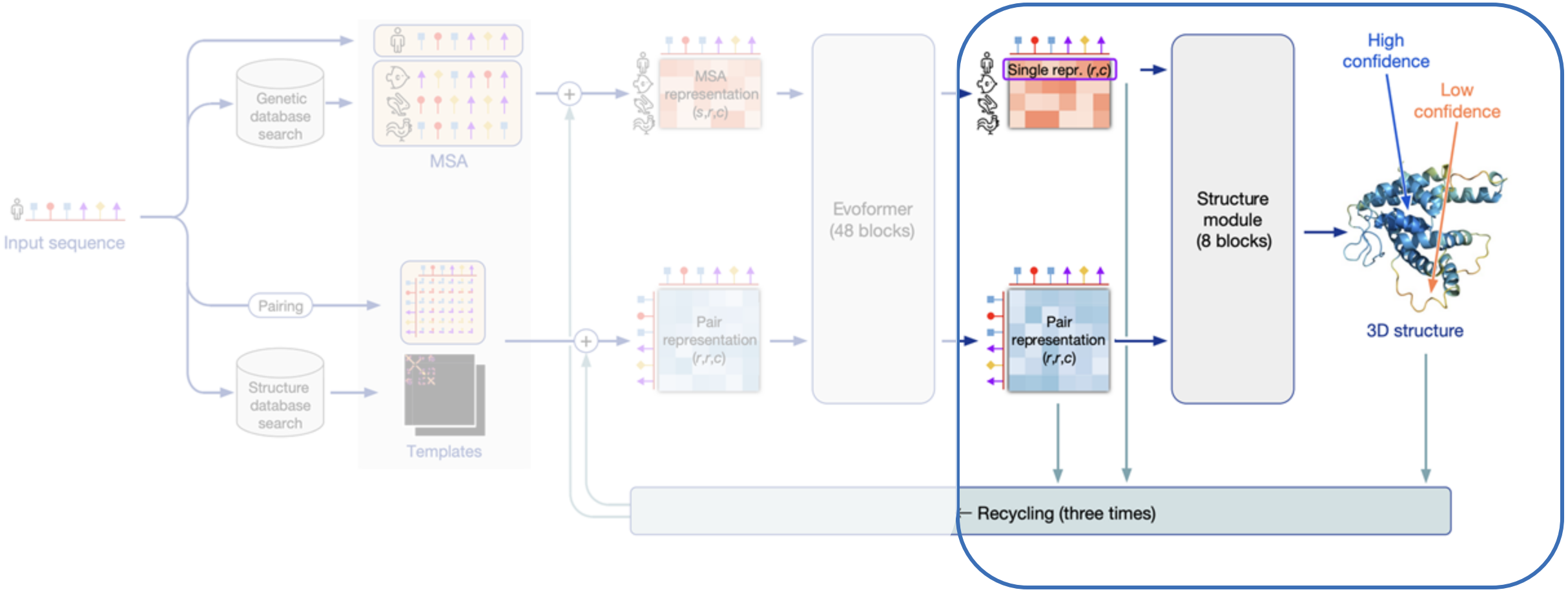
The Evoformer interprets and updates both the MSA and the pair representations. The important aspect of this network is the continuous flow of information between the MSA and the pair representations. This enables reasoning about spatial and evolutionary relationships, which refines the structural hypothesis.

Google DeepMind

PDBe Protein Data Bank in Europe EMBL-EBI

Jumper, J et al. Highly accurate protein structure prediction with AlphaFold. Nature (2021).

Structure module

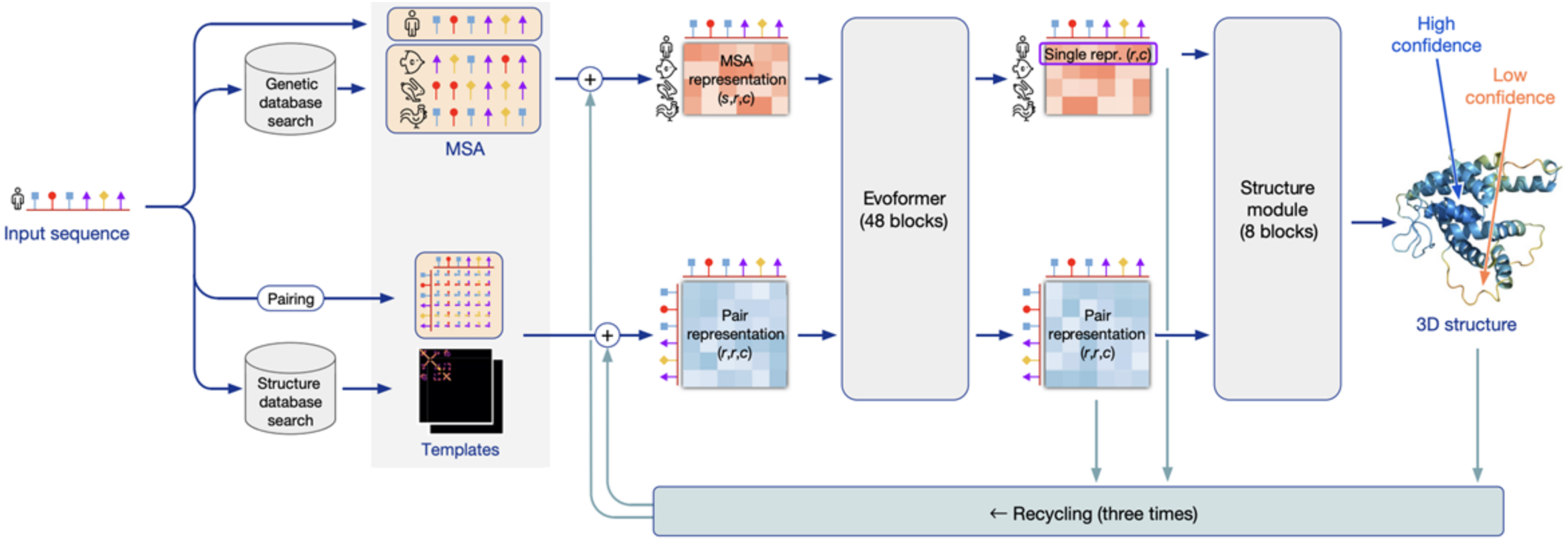


Google DeepMind

PDBe Protein Data Bank in Europe EMBL-EBI

Jumper, J et al. Highly accurate protein structure prediction with AlphaFold. Nature (2021).

Recycling



Google DeepMind

PDBe Protein Data Bank in Europe EMBL-EBI

Jumper, J et al. Highly accurate protein structure prediction with AlphaFold. Nature (2021).

AlphaFold structures closely resemble crystal structures

- Structural biology experiments demonstrate that AlphaFold structures (or well-defined parts of the predicted structures, like protein domains) **work well as search models for molecular replacement** (MR) in X-ray crystallography.
  - Molecular replacement works well if RMSD < 1-1.5 Å for at least 30% of the atoms.
  - Availability of high-accuracy models is then definitely critical for successful structure solution.

The X-ray crystallography phase problem solved thanks to *AlphaFold* and *RoseTTAFold* models: a case-study report

I. Barbarin-Bocahu and M. Graille

The breakthrough recently made in protein structure prediction by deep-learning programs such as *AlphaFold* and *RoseTTAFold* will certainly revolutionize biology over the coming decades. The scientific community is only starting to appreciate the various applications, benefits and limitations of these protein models. Yet, after the first thrills due to this revolution, it is important to evaluate the impact of the proposed models and their overall quality to avoid the misinterpretation or overinterpretation of these models by biologists. One of the first applications of these models is in solving the `phase problem' encountered in X-ray crystallography in calculating electron-density maps from diffraction data. Indeed, the most frequently used technique to derive electron-density maps is molecular replacement. As this technique relies on knowledge of the structure of a protein that shares strong structural similarity with the studied protein, the availability of high-accuracy models is then definitely critical for successful structure solution. After the collection of a 2.45 Å resolution data set, we struggled for two years in trying to solve the crystal structure of a protein involved in the nonsense-mediated mRNA decay pathway, an mRNA quality-control pathway dedicated to the elimination of eukaryotic mRNAs harboring premature stop codons. We used different methods (isomorphous replacement, anomalous diffraction and molecular replacement) to determine this structure, but all failed until we straightforwardly succeeded thanks to both *AlphaFold* and *RoseTTAFold* models. Here, we describe how these new models helped us to solve this structure and conclude that in our case the *AlphaFold* model largely outcompetes the other models. We also discuss the importance of search-model generation for successful molecular replacement.

Implications of *AlphaFold*2 for crystallographic phasing by molecular replacement

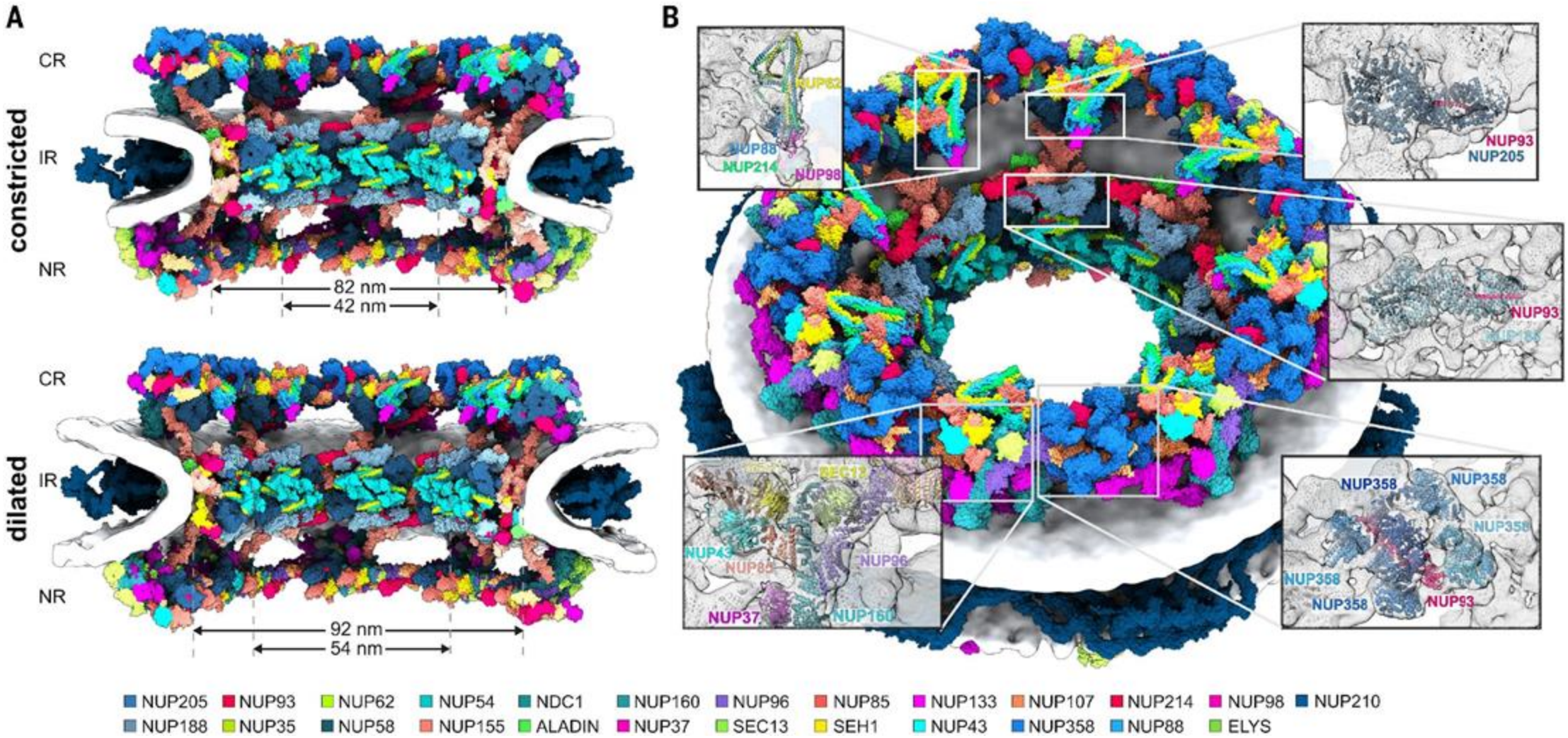
Airlie J. McCoy,[a]* Massimo D. Sammito[a] and Randy J. Read[a]

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Edited by D. J. Rigden, University of Liverpool, United Kingdom (Received 18 May 2021; accepted 13 November 2021)

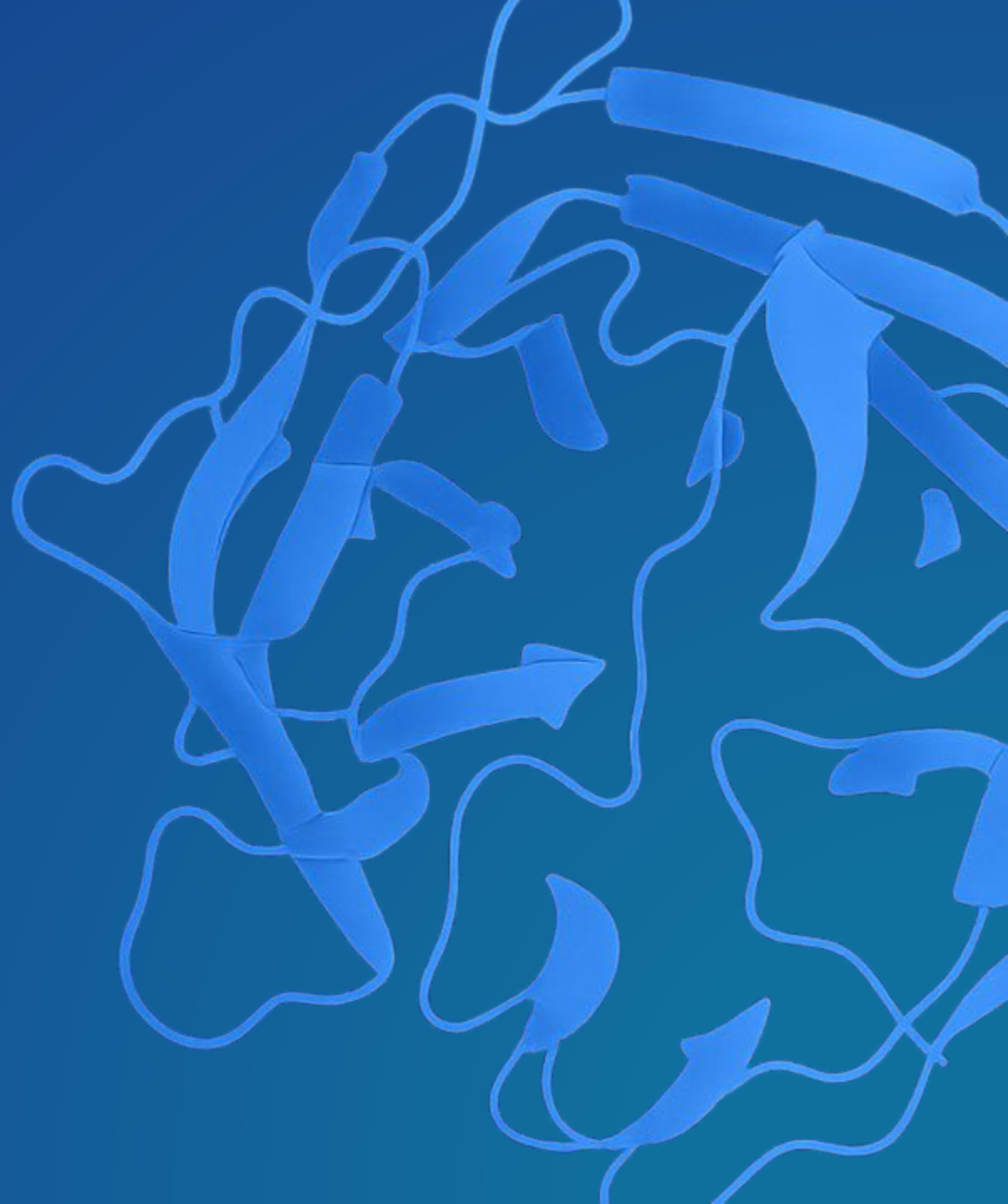
The *AlphaFold*2 results in the 14th edition of Critical Assessment of Structure Prediction (CASP14) showed that accurate (low root-mean-square deviation) *in silico* models of protein structure domains are on the horizon, whether or not the protein is related to known structures through high-coverage sequence similarity. As highly accurate models become available, generated by harnessing the power of correlated mutations and deep learning, one of the aspects of structural biology to be impacted will be methods of phasing in crystallography. Here, the data from CASP14 are used to explore the prospects for changes in phasing methods, and in particular to explore the prospects for molecular-replacement phasing using *in silico* models.

AlphaFold to solve huge complexes



Shyamal Mosalaganti et al. ,AI-based structure prediction empowers integrative structural analysis of human nuclear pores.Science (2022)

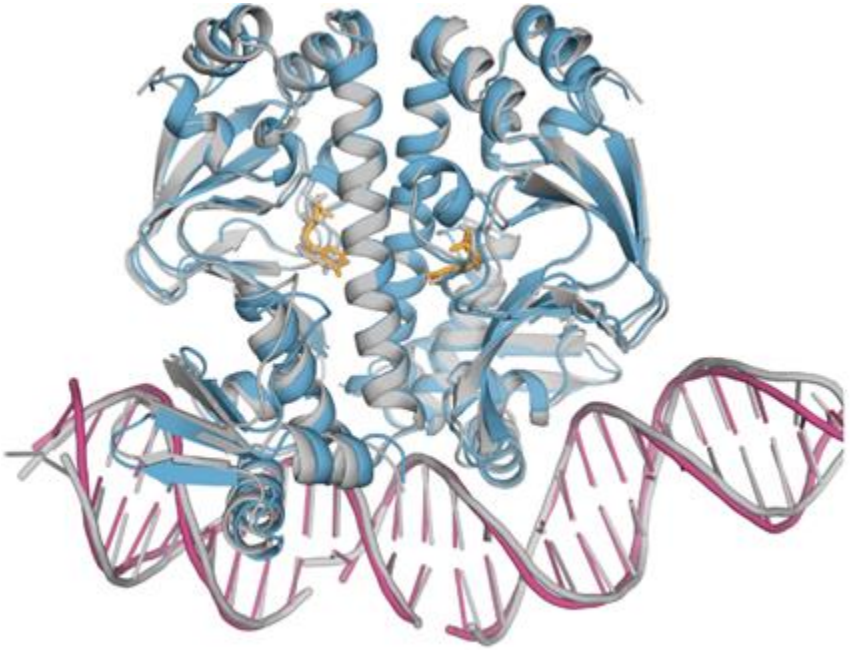
Introducing AlphaFold3 & AlphaFold Server



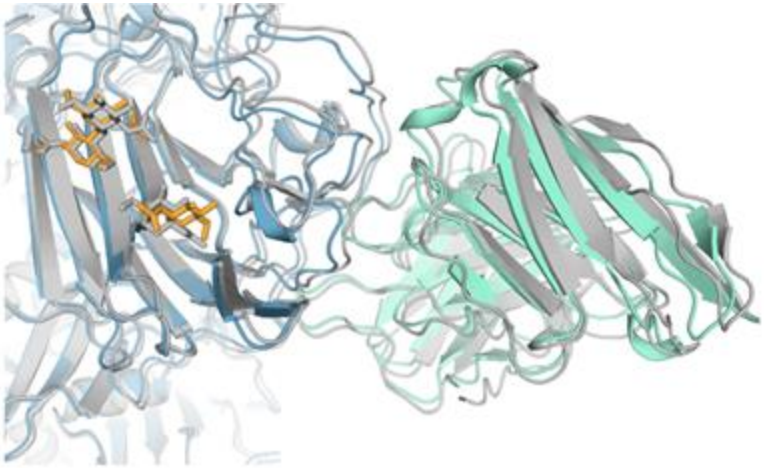
AlphaFold 3 (AF3)

- ✅ Proteins
- ✅ RNA
- ✅ DNA
- ✅ Small molecules
- ✅ Ions
- ✅ Modified residues
- ✅ Glycosylation
- ✅ …
- ❌ Water
- ❌ Hydrogen

Example structures predicted using AF3



Bacterial CRP/FNR family transcriptional regulator protein bound to DNA and cGMP (PDB 7PZB; full-complex LDDT, 82.8; global distance test (GDT) 90.1).

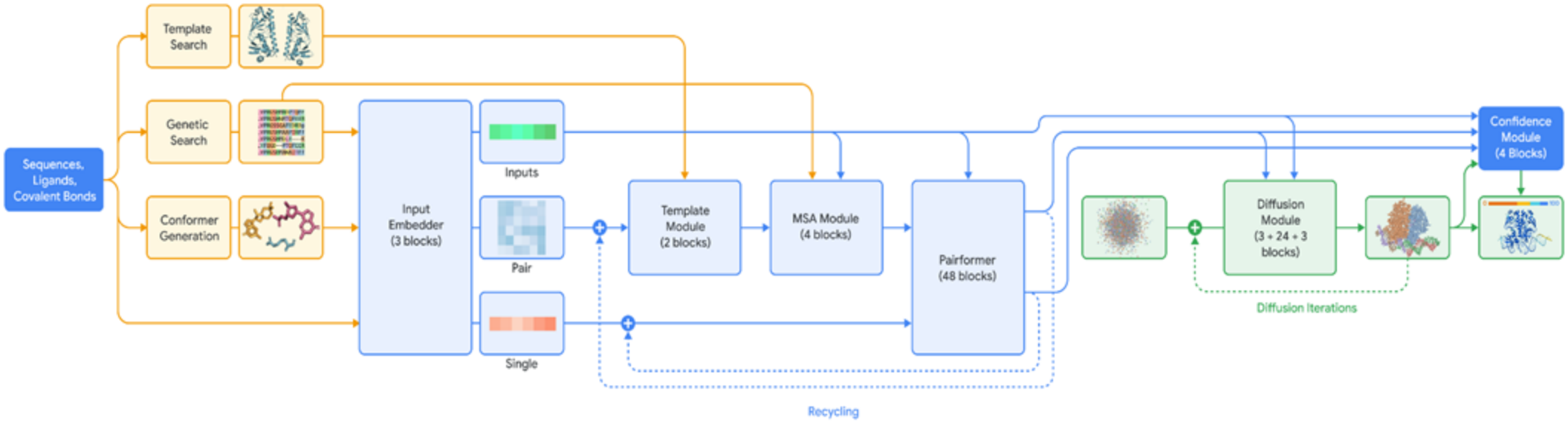


Human coronavirus OC43 spike protein, 4,665 residues, heavily glycosylated and bound by neutralizing antibodies (PDB 7PNM; full-complex LDDT: 83.0; GDT: 83.1)

Abramson, J., Adler, J., Dunger, J. et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature 630, 493–500 (2024).

AlphaFold 3 architecture

AlphaFold 3 predicts raw atom coordinates using a diffusion module, making it a “generative” ML model. The diffusion module was trained to receive “noised” atomic coordinates, and to use them to predict the correct coordinates



Abramson, J., Adler, J., Dunger, J. et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature 630, 493–500 (2024).

Summing up

- Proteins are fundamental to life (and biotechnology)
- Knowledge of structure at atomic level facilitates understanding of function and how this might be enhanced
- X-ray crystallography bedrock of experimental structure solution
- CryoEM increasingly popular
- AlphaFold2 step change in computational prediction of structure

